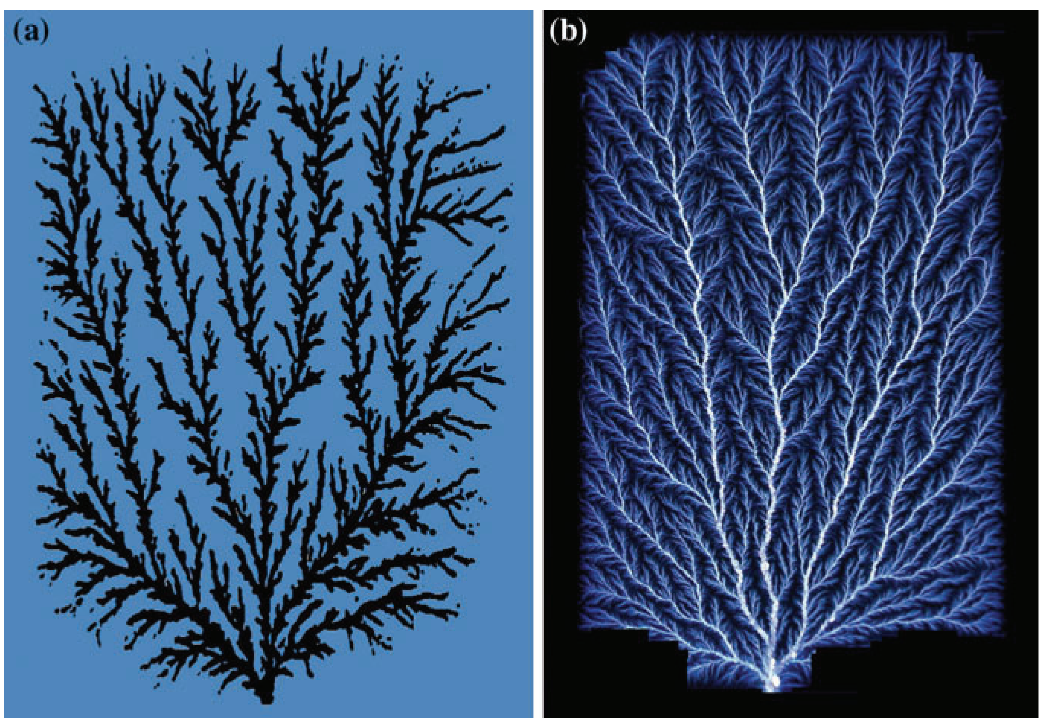
Chapter 6
Interactions of Charged Particles with Matter

In this chapter we discuss interactions of charged particle radiation with matter. A charged particle is surrounded by its Coulomb electric field that interacts with orbital electrons and the nucleus of all atoms it encounters, as it penetrates into matter. Charged particle interactions with orbital electrons of the absorber result in collision loss, interactions with nuclei of the absorber result in radiation loss. The energy transfer from the charged particle to matter in each individual atomic interaction is generally small, so that the particle undergoes a large number of interactions before its kinetic energy is spent.



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Stopping power is the parameter used to describe the gradual loss of energy of the charged particle, as it penetrates into an absorbing medium. Two classes of stopping power are known: *collision stopping power* that results from charged particle interaction with orbital electrons of the absorber and *radiation stopping power* that results from charged particle interaction with nuclei of the absorber.

Stopping powers play an important role in radiation dosimetry. They depend on the properties of the charged particle such as its mass, charge, velocity and energy as well as on the properties of the absorbing medium such as its density and atomic number. In addition to stopping powers, other parameters of charged particle interaction with matter, such as the range, energy transfer, mean ionization potential, and radiation yield, are also discussed in this chapter.

6.1 General Aspects of Energy Transfer from Charged Particle to Medium

The discovery of energetic charged particle emission from radioactive materials in 1896 stimulated interest not only in the origin of the emitted particles but also in how they were slowed down as they traversed matter. The theory of stopping power played an important role in the development of atomic and nuclear models starting with the α-particle scattering studies of *Hans Geiger*, *Ernest Marsden* and *Ernest Rutherford* in 1908 and the classical stopping power theory developed by *Niels Bohr* in 1913, and culminating with the quantum mechanical and relativistic theory of stopping power proposed by *Hans Bethe* in the 1930s and refined by *Ugo Fano* in the 1960s. More recent developments introduced several additional secondary correction factors to increase the accuracy of theoretical stopping power expressions; however, the main theoretical foundations that early workers enunciated decades ago are still valid today.

As a charged particle travels through an absorber, it experiences Coulomb interactions with the nuclei and orbital electrons of absorber atoms. These interactions can be divided into three categories depending on the size of the classical impact parameter b of the charged particle trajectory compared to the classical atomic radius a of the absorber atom with which the charged particle interacts:

1. Coulomb force interaction of the charged particle with the external nuclear field of the absorber atom for $b \ll a$ (bremsstrahlung production).
2. Coulomb force interaction of the charged particle with orbital electron of the absorber atom for $b \approx a$ (hard collision).
3. Coulomb force interaction of the charged particle with orbital electron of the absorber atom for $b \gg a$ (soft collision).

Radiation collision, hard collision, and soft collision are shown schematically in Fig. 6.1, with b the impact parameter of the particle trajectory and a the atomic radius of the absorber atom.

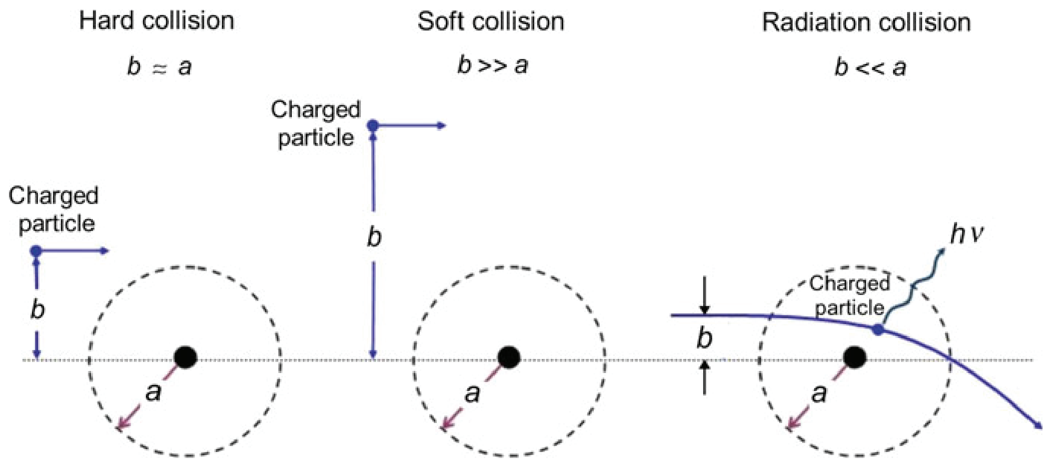


Fig. 6.1 Three different types of collision of a charged particle with an atom, depending on the relative size of the impact parameter b and atomic radius a. Hard (close) collision for $b \approx a$; soft (distant) collision for $b \gg a$; and radiation collision for $b \ll a$

6.1.1 Charged Particle Interaction with Coulomb Field of the Nucleus (Radiation Collision)

When the impact parameter b of a charged particle is much smaller than the radius a of the absorber atom (i.e., $b \ll a$), the charged particle interacts mainly with the nucleus and undergoes either elastic or inelastic scattering possibly accompanied with a change in direction of motion.

The vast majority of these interactions are elastic so that the particle is scattered by the nucleus but loses only an insignificant amount of its kinetic energy to satisfy the conservation of momentum requirement. However, a small percentage of the scattering interactions are inelastic and may result in significant energy loss for the charged particle accompanied by emission of x-ray photons. This type of interaction is called *bremsstrahlung collision*. At a given particle acceleration, the probability for this type of interaction is inversely proportional to the square of the mass of the charged particle, making the bremsstrahlung production for charged particles other then electrons and positrons essentially negligible.

6.1.2 Hard (Close) Collision

When the impact parameter b of a charged particle trajectory is of the order of the radius a of the absorber atom (i.e., $b \approx a$), the charged particle may have a direct Coulomb impact interaction with a single atomic orbital electron and transfer to it a significant amount of energy. The interaction is referred to as *hard* or *close collision*.

The orbital electron leaves the atom as a δ-ray, and is energetic enough to undergo its own Coulomb interactions with absorber atoms. The maximum possible energy transfer from a charged particle to an orbital electron (δ-ray) is discussed in detail in

Sect. 5.3. The number of hard collisions experienced by a charged particle moving in an absorber is generally small; however, the energy transfers associated with hard collisions are relatively large, so that the particle loses roughly 50% of its kinetic energy through hard collisions.

The theories that govern hard collisions depend strongly on the characteristics of charged particles and generally assume that the orbital electron (δ-ray) released through a hard collision is free before and after the interaction, since the kinetic energy transferred to it from the charged particle is much larger than its atomic binding energy.

6.1.3 Soft (Distant) Collision

When the impact parameter b of the charged particle trajectory is much larger than the radius a of the absorber atom (i.e., $b \gg a$), the charged particle interacts with the whole atom and the whole atomic complement of bound electrons. The interaction is called a *soft* or *distant collision*. The energy transfer from the charged particle to a given bound electron is very small; however, the number of these interactions is large, so that approximately 50% of energy loss by a charged particle occurs through these small-energy-transfer interactions that may cause atomic polarization, excitation or ionization through removal of a valence electron. In the energy region of soft collisions the expressions derived with a given theory are valid for all types of charged particles including electrons and positrons.

6.2 General Aspects of Stopping Power

During its motion through an absorbing medium a charged particle experiences a large number of interactions before its kinetic energy is expended. In each interaction the charged particle's path may be altered (*elastic* or *inelastic scattering*) and it may lose some of its kinetic energy that will be transferred to the medium (*collision loss*) or to photons (*radiation loss*). Each of these possible interactions between the charged particle and orbital electrons or the nucleus of the absorber atoms is characterized by a specific cross section (probability) σ for the particular interaction. The energy loss of the charged particle propagating through an absorber depends on the characteristics of the particle as well as the absorber.

The rate of energy loss (typically expressed in MeV) per unit of path length (typically expressed in cm) by a charged particle in an absorbing medium is called the linear stopping power $(-\mathrm{d}E/\mathrm{d}x)$. Dividing the linear stopping power by the density ρ of the absorber results in the mass stopping power S given in units of $\mathrm{MeV{\cdot}cm^2{\cdot}g^{-1}}$. The stopping power is a property of the material in which a charged particle propagates.

In general, the average energy loss per unit path length $-\mathrm{d}E/\mathrm{d}\ell$ experienced by the heavy particle is calculated by multiplying the cross section for a given energy loss σ_{ni} by the energy loss ΔE_{ni} and a summation over all possible individual collisions i

$$-\frac{\mathrm{d}E}{\mathrm{d}\ell} = \sum_i N_i \sum_n \Delta E_{ni}\sigma_{ni}, \tag{6.1}$$

where N_i is the density of atoms i that can be expressed:

1. Either in number of atoms per unit volume resulting in $-\mathrm{d}E/\mathrm{d}\ell$ referred to as the linear stopping power $-\mathrm{d}E/\mathrm{d}x$ and representing energy loss per unit distance traversed in the absorber. The typical units of linear stopping power are MeV/cm or less common keV/μm.
2. Or in number of atoms per unit mass resulting in $-\mathrm{d}E/\mathrm{d}\ell$ referred to as mass stopping power $S = -(1/\rho)\,\mathrm{d}E/\mathrm{d}x$ and representing energy loss per g/cm^2 of material traversed in the absorber. The typical unit of mass stopping power is MeV·cm^2·g^{-1}.

With regard to charged particle interaction, two types of stopping power are known:

1. *Radiation stopping power* (also called nuclear stopping power) resulting from charged particle Coulomb interaction with the nuclei of the absorber. Only light charged particles (electrons and positrons) experience appreciable energy loss through these interactions that are usually referred to as bremsstrahlung interactions. For heavy charged particles (protons, α-particles, etc.) the radiation (bremsstrahlung) loss is negligible in comparison with the collision loss.
2. *Collision stopping power* (also called ionization or electronic stopping power) resulting from charged particle Coulomb interactions with orbital electrons of the absorber. Both heavy and light charged particles experience these interactions that result in energy transfer from the charged particle to orbital electrons through impact excitation and ionization of absorber atoms.

The total stopping power S_{tot} for a charged particle of kinetic energy E_{K} traveling through an absorber of atomic number Z is in general the sum of the radiation (nuclear) stopping power S_{rad} and collision (electronic) stopping power S_{col}, i.e.,

$$S_{\mathrm{tot}} = S_{\mathrm{rad}} + S_{\mathrm{col}}. \tag{6.2}$$

The collision stopping power S_{col} is further subdivided into two components: the soft (distant) collision stopping power $S_{\mathrm{col}}^{\mathrm{soft}}$ and the hard (close) collision stopping power $S_{\mathrm{col}}^{\mathrm{hard}}$

$$S_{\mathrm{col}} = S_{\mathrm{col}}^{\mathrm{soft}} + S_{\mathrm{col}}^{\mathrm{hard}} \tag{6.3}$$

The total stopping power is thus in general terms expressed as the following sum

$$S_{\mathrm{tot}} = S_{\mathrm{rad}} + S_{\mathrm{col}} = S_{\mathrm{rad}} + S_{\mathrm{col}}^{\mathrm{soft}} + S_{\mathrm{col}}^{\mathrm{hard}} \tag{6.4}$$

6.3 Radiation (Nuclear) Stopping Power

As shown by the Larmor relationship (4.18), any time a charged particle is accelerated or decelerated part of its kinetic energy is emitted in the form of bremsstrahlung photons. The rate of bremsstrahlung energy dissipation is proportional to a^2 (the square of the charged particle acceleration a) which in turn is proportional to $(zZ/m)^2$ with z and m the atomic number and mass, respectively, of the radiating charged particle, and Z the atomic number of the absorber target.

The bremsstrahlung intensity is thus linearly proportional to $(zZ)^2$ and inversely proportional to m^2. As a consequence of the relatively large mass of heavy charged particles, the bremsstrahlung yield produced by heavy charged particles such as protons and α-particles in comparison with electrons and positrons is insignificant and generally ignored.

Hans Bethe and *Walter Heitler* have shown in 1930s that the cross section for emission of bremsstrahlung σ_{rad} has the same form in classical and quantum theory and is proportional to

$$\sigma_{\text{rad}} \propto \alpha r_{\text{e}}^2 Z^2 \left(\text{cm}^2/\text{nucleus}\right), \tag{6.5}$$

where

α is the fine structure constant $\left[e^2/\left(4\pi\varepsilon_0\hbar c\right) = 1/137\right]$.
r_{e} is the classical electron radius $\left[e^2/\left(4\pi\varepsilon_0 m_{\text{e}}c^2\right) = 2.818\ \text{fm}\right]$.
Z is the atomic number of the absorber target.

Table 6.1 provides expressions for σ_{rad} for various regions of incident electron kinetic energy $(E_{\text{K}})_0$ from the classical region where $(E_{\text{K}})_0 \ll m_{\text{e}}c^2$ all the way to the extreme relativistic region where $(E_{\text{K}})_0 \gg m_{\text{e}}c^2/\alpha$.

The rate of bremsstrahlung production by light charged particles (electrons and positrons) traveling through an absorber is generally expressed by the mass radiation stopping power S_{rad} (in MeV·cm^2·g^{-1}) given as follows

$$S_{\text{rad}} = N_{\text{a}}\,\sigma_{\text{rad}}\,E_{\text{i}}, \tag{6.6}$$

where

N_{a} is the atomic density, i.e., number of atoms per unit mass: $N_{\text{a}} = N/m = N_{\text{A}}/A$.
σ_{rad} is the total cross section for bremsstrahlung production given for various energy regions in Table 6.1.
E_{i} is the initial total energy of the light charged particle, i.e., $E_{\text{i}} = (E_{\text{K}})_0 + m_{\text{e}}c^2$.
$(E_{\text{K}})_0$ is the initial kinetic energy of the light charged particle.

Table 6.1 Total cross section for bremsstrahlung production and parameter B_{rad} for various ranges of electron kinetic energies

Energy range	σ_{rad} $(cm^2/nucleon)$	$B_{rad} = \sigma_{rad}/\left(\alpha r_e^2 Z^2\right)$	
Non-relativistic $(E_K)_0 \ll m_e c^2$	$\frac{16}{3}\alpha r_e^2 Z^2$	$\frac{16}{3}$	(6.8)
Relativistic $(E_K)_0 \approx m_e c^2$	Complicated power series	–	(6.9)
High-relativistic $m_e c^2 \ll (E_K)_0 \ll \frac{m_e c^2}{\alpha Z^{1/3}}$	$8 r_e^2 Z^2 \left[\ln\left(\frac{E_i}{m_e c^2}\right) - \frac{1}{6}\right]$	$8\left[\ln\left(\frac{E_i}{m_e c^2}\right) - \frac{1}{6}\right]$	(6.10)
Extreme relativistic $(E_K)_0 \gg \frac{m_e c^2}{}\alpha Z^{1/3}$	$4\alpha r_e^2 Z^2\left[\ln\frac{183}{Z^{1/3}} + \frac{1}{18}\right]$	$4\left[\ln\frac{183}{Z^{1/3}} + \frac{1}{18}\right]$	(6.11)

Inserting σ_{rad} from Table 6.1 into (6.6) we obtain the following expression for S_{rad}

$$S_{rad} = \alpha\, r_e^2 Z^2 \frac{N_A}{A} E_i B_{rad}, \tag{6.7}$$

where B_{rad} is a slowly varying function of Z and E_i, also given in Table 6.1 and determined from $\sigma_{rad}/\left(\alpha\, r_e^2 Z^2\right)$. As shown in Table 6.2, the parameter B_{rad} has a value of $\frac{16}{3}$ for light charged particles in the non-relativistic energy range $(E_K)_0 \ll m_e c^2$; about 6 at $(E_K)_0 = 1$ MeV; 12 at $(E_K)_0 = 10$ MeV; and 15 at $(E_K)_0 = 100$ MeV. *Hans Bethe* and *Walter Heitler* derived (6.7) theoretically. *Martin Berger* and *Stephen Seltzer* have provided extensive tables of S_{rad} for a wide range of absorbing materials. As indicated in (6.7), the mass radiation stopping power S_{rad} is proportional to:

1. $\left(N_A Z^2/A\right)$ indicating a proportionality with the atomic number of the absorber Z by virtue of $Z/A \approx 0.5$ for all elements with the exception of hydrogen. The higher is the atomic number Z of the absorber, the larger is the radiation stopping power S_{rad} and the larger is the radiation yield.
2. Total energy E_i [(or kinetic energy $(E_K)_0$ for $(E_K)_0 \gg m_e c^2$] of the light charged particle.
3. Parameter B_{rad} which is a slowly varying function of light charged particle total energy E_i and absorber atomic number Z, as shown in Table 6.2.

Figure 6.2 shows the mass radiation stopping power S_{rad} for electrons in water, aluminum, and lead based on tabulated data obtained from the National Institute of Standards and Technology (NIST). The S_{rad} data are shown with heavy solid curves,

Table 6.2 Parameter B_{rad} for various initial kinetic energies of light charged particles

Kinetic energy	Classical	1 MeV	10 MeV	100 MeV
B_{rad}	~5.3	~6	~10	~15

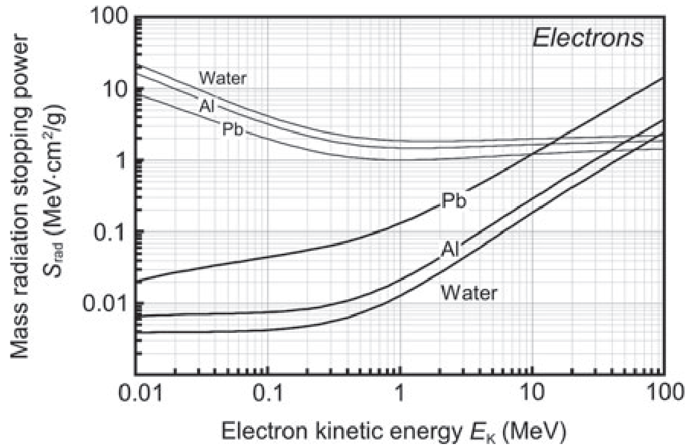


Fig. 6.2 Mass radiation stopping power S_{rad} for electrons in water, aluminum and lead shown with heavy solid curves against the electron kinetic energy E_{K}. Mass collision stopping powers S_{col}, discussed in Sect. 6.5, for the same materials are shown with light solid curves for comparison. Data were obtained from the NIST

mass collision stopping powers S_{col} (discussed in Sect. 6.5) are shown with light curves for comparison. The radiation stopping power S_{rad} clearly shows an approximate proportionality: (1) to the atomic number Z of the absorber at a given initial kinetic energy of the light charged particle and (2) to initial kinetic energy $(E_{\text{K}})_0$ of the light charged particle for a given absorber material.

6.4 Collision (Electronic) Stopping Power for Heavy Charged Particles

Energy transfer from energetic heavy charged particles to a medium (absorber) they traverse occurs mainly through Coulomb interactions of the charged particles with orbital electrons of the absorber atoms (collision or electronic loss); inelastic Coulomb interactions between heavy charged particles and nuclei of absorber atoms (radiation loss) are negligible and thus ignored.

Two different approaches were developed to describe a heavy charged particle energy loss to orbital electrons of absorber atoms:

1. Bohr's approach (1913) is in the realm of classical physics and is based on the concept of impact parameter between the particle's trajectory and the absorber nucleus.
2. Bethe's approach (1931) is in the realm of quantum mechanics and relativistic physics, and assumes that the momentum transfer related to the particle's energy loss is quantized.

The basic theories dealing with collision loss of energetic heavy charged particles in absorbing media make the following assumptions:

1. The energetic charged particle is moving through the absorber much faster than the orbital electrons of absorber atoms.
2. The energetic charged particle is much heavier than the energy-absorbing orbital electrons.

3. The energetic charged particle interacts with absorber atoms only through electromagnetic forces; nuclear reactions between the particle and absorber nuclei are not considered.
4. The energetic charged particle loses energy through interactions with orbital electrons of the absorber; elastic and inelastic interactions between the heavy charged particle and nuclei of the absorber are negligible.

6.4.1 Momentum and Energy Transfer from Heavy Charged Particle to Orbital Electron

In 1913 *Niels Bohr* was first to propose a theory describing a heavy charged particle energy loss in an absorbing medium. His classical derivation of the mass collision stopping power S_{col} for a heavy charged particle, such as a proton, is based on the calculation of the momentum change (impulse) Δp of the heavy charged particle having a Coulomb interaction with an orbital electron.

The Coulomb interaction between the heavy charged particle (charge ze and mass M) and the orbital electron (charge e and mass m_e) is shown schematically in Fig. 6.3. The situation here seems similar to that depicted in Fig. 2.3 for Rutherford scattering between an α-particle with mass m_α and gold nucleus with mass M. We must note, however, that in Rutherford scattering $m_\alpha \ll M$, where m_α is the mass of the projectile (α-particle) and M is the mass of the target (nucleus) while the case here is reversed as we have a heavy charged particle with mass M in translational motion and interacting with a stationary orbital electron with mass m_e where $M \gg m_e$. Assuming that the heavy particle is positively charged, the orbital electron is located in the inner focus of the hyperbolic trajectory that, in principle, the heavy charged particle follows. In Rutherford scattering, on the other hand, the α-particle and the positively charged nucleus repel one another and the nucleus is located in the outer focus of the hyperbolic trajectory that the α-particle follows.

The momentum transfer (impulse) Δp is along a line that bisects the angle $\pi - \theta$, as indicated in Fig. 6.3, and the magnitude of Δp is calculated in a similar manner to that used for Rutherford scattering in Sect. 2.3.3

$$\Delta p = \int F_{\Delta p}\,\mathrm{d}t = \int_{-\infty}^{\infty} F_{\mathrm{Coul}} \cos\phi\,\mathrm{d}t. \tag{6.12}$$

The magnitude of the Coulomb force F_{Coul} between the heavy particle of charge ze and the orbital electron of charge $-e$ is

$$F_{\mathrm{Coul}} = \frac{ze^2}{4\pi\varepsilon_o}\frac{1}{r^2}, \tag{6.13}$$

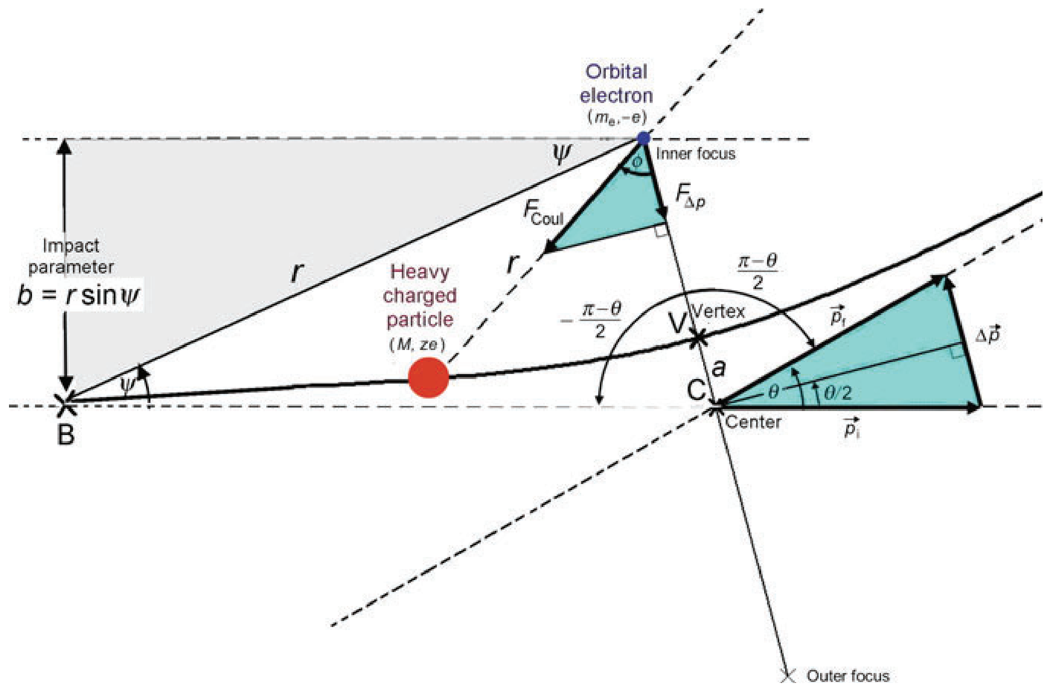


Fig. 6.3 Schematic diagram of a collision between a positively charged heavy particle with mass M and an orbital electron with mass m_e. Since $M \gg m_e$, the scattering angle $\theta \approx 0°$. The scattering angle is shown larger than 0° to highlight the principles of the Coulomb collision and aid in the derivation of Δp. The electron is in the inner focus of the hyperbola because of the attractive Coulomb force between the positive heavy charged particle and the negative orbital electron. Point B is for the charged particle at a very large distance from the orbital electron; point V is for the charged particle at the vertex V of the hyperbolic trajectory

Incorporating the expression for the Coulomb force (6.13) into (6.12), the momentum transfer Δp can now be written as

$$\Delta p = \frac{ze^2}{4\pi\varepsilon_o} \int_{-\frac{\pi-\theta}{2}}^{\frac{\pi-\theta}{2}} \frac{\cos\phi}{r^2}\frac{dt}{d\phi} d\phi = \frac{ze^2}{4\pi\varepsilon_o} \int_{-\frac{\pi-\theta}{2}}^{\frac{\pi-\theta}{2}} \frac{\cos\phi}{r^2\omega} d\phi \tag{6.14}$$

where ϕ is the angle between the radius vector r and the bisector (axis) of the hyperbola, as shown in Fig. 6.3, and ω is the angular frequency defined as $\omega = \frac{d\phi}{dt} = \frac{\upsilon}{r}$ with υ the initial velocity of the incident charged particle.

The angular momentum L for the elastic collision process between a heavy charged particle M and an electron m_e is in general expressed as

$$L = |\mathbf{L}|\, |\mathbf{r} \times \mathbf{M}\boldsymbol{\upsilon}| = rM\upsilon \sin\psi = Mr^2\omega \sin\psi \tag{6.15}$$

where υ and $\boldsymbol{\upsilon}$ are the incident velocity and velocity vector, respectively, of the charged particle and ψ, as shown in Fig. 6.3, is the angle between the radius vector $\mathbf{r}$ and vector velocity $\boldsymbol{\upsilon}$ (see also Fig. 2.4 and Sect. 2.33).

We now use conservation of angular momentum L to express ωr^2 of (6.14) in terms of impact parameter b by determining L for two special points (B and V) on the hyperbola: point B is at a large distance (∞) from the center of the hyperbola and point V is at the vertex of the hyperbola. Note: at point B we have $b = r\sin\psi$ and at point V the angle ψ between $\mathbf{r}$ and $\boldsymbol{\upsilon}$ is 90° and $\sin\psi = 1$. We can thus express L_B and L_V as follows

$$L_\mathrm{B} = rM\upsilon\sin\psi = M\upsilon b \qquad \text{and} \qquad L_\mathrm{v} = rM\upsilon = M\omega r^2 \tag{6.16}$$

from where it follows that ωr^2 of (6.14) is, as a result of angular momentum conservation ($L = L_\mathrm{B} = L_\mathrm{v}$), equal to υb.

The momentum transfer Δp of (6.14) can now be simplified as follows

$$\Delta p = \frac{ze^2}{4\pi\varepsilon_\mathrm{o}}\frac{1}{\upsilon b}\int\limits_{-\frac{\pi-\theta}{2}}^{\frac{\pi-\theta}{2}} \cos\phi\,\mathrm{d}\phi = \frac{ze^2}{4\pi\varepsilon_\mathrm{o}}\frac{1}{\upsilon b}\{\sin\phi\}_{-(\pi-\theta)/2}^{(\pi-\theta)/2} = 2\frac{ze^2}{4\pi\varepsilon_\mathrm{o}}\frac{1}{\upsilon b}\cos\frac{\theta}{2}. \tag{6.17}$$

Equation (6.17) is identical to the Rutherford expression for Δp in (2.21). However, in the case of a heavy charged particle (ze) interacting with a stationary orbital electron (e), the scattering angle $\theta \approx 0$ and $\cos\left(\frac{1}{2}\theta\right) \approx 1$, resulting in the following simplified expression for Δp

$$\Delta p = 2\frac{ze^2}{4\pi\varepsilon_\mathrm{o}}\frac{1}{\upsilon b}. \tag{6.18}$$

The energy transferred to the orbital electron from the heavy charged particle for a single interaction with an impact parameter b is

$$\Delta E(b) = \frac{(\Delta p)^2}{2m_\mathrm{e}} = 2\left(\frac{e^2}{4\pi\varepsilon_\mathrm{o}}\right)^2\frac{z^2}{m_\mathrm{e}\upsilon^2 b^2}, \tag{6.19}$$

using the classical expression between kinetic energy $E_\mathrm{K} = \frac{1}{2}m_\mathrm{e}\upsilon^2$ and momentum $p = m_\mathrm{e}\upsilon$ expressed as $E_\mathrm{K} = \frac{1}{2}p^2/m_\mathrm{e}$. Note that in (6.19) m_e is the rest mass of the electron (target) and υ is the velocity of the heavy charged particle (projectile), so that $\frac{1}{2}m_\mathrm{e}\upsilon^2$ should not be misconstrued for the kinetic energy of the projectile. From (6.18) and (6.19) we express the impact parameter b as

$$b = 2\left(\frac{e^2}{4\pi\varepsilon_0}\right)\frac{z}{\upsilon\Delta p} = \sqrt{\frac{2}{m_\mathrm{e}}}\left(\frac{e^2}{4\pi\varepsilon_0}\right)\frac{z}{\upsilon[\Delta E(b)]^{1/2}}. \tag{6.20}$$

Following (6.1) we set up a summation over all possible collisions to determine the mass collision stopping power for heavy charged particles

$$\begin{aligned} S_{\mathrm{col}} &= -\frac{1}{\rho}\frac{\mathrm{d}E}{\mathrm{d}x} = N_{\mathrm{e}}\int \Delta E(b)\mathrm{d}\sigma_{\mathrm{col}} = N_{\mathrm{e}}\int \Delta E(b)\frac{\mathrm{d}\sigma_{\mathrm{col}}}{\mathrm{d}b}\,\mathrm{d}b \\ &= \int \Delta E(b)\frac{\mathrm{d}\sigma_{\mathrm{col}}}{\mathrm{d}(\Delta E)}\mathrm{d}\,(\Delta E), \end{aligned} \tag{6.21}$$

where N_{e} is the electron density (number of electrons per unit mass) of the absorber ($N_{\mathrm{e}} = ZN_{\mathrm{A}}/A$), and $\mathrm{d}\sigma_{\mathrm{col}}/\mathrm{d}b$ as well as $\mathrm{d}\sigma_{\mathrm{col}}/\mathrm{d}(\Delta E)$ are differential cross sections for the collision interaction.

The mass collision stopping power S_{col} is calculated by integrating $\Delta E(b)$ of (6.21) over all possible impact parameters b ranging from $b_{\min}$ to $b_{\max}$ or over all possible energy transfers $\Delta E(b)$ ranging from $\Delta E_{\min}$ to $\Delta E_{\max}$. Intuitively we might have considered integration in (6.21) over all possible impact parameters b from 0 to ∞ or over all possible energy transfers ΔE from 0 to the kinetic energy of the incident charged particle $(E_{\mathrm{K}})_{\mathrm{i}}$; however, we must account for two physical limitations affecting the energy transfer from a heavy charged particle to an orbital electron which is bound to the atomic nucleus:

1. *Minimum possible energy transfer* $\Delta E_{\min}$ is governed by the ionization and excitation potential of orbital electrons of absorber atoms resulting in a minimum possible energy transfer $\Delta E_{\min}$ below which energy transfer becomes impossible. Since energy transfer ΔE and impact parameter b are inversely proportional, as shown in (6.19), $\Delta E_{\min}$ corresponds to the maximum impact parameter $b_{\max}$ beyond which energy transfer becomes impossible.
2. *Maximum possible energy transfer* $\Delta E_{\max}$ in a head-on collision between the heavy charged particle and an orbital electron is governed by the masses of the two interacting particles and is significantly lower than the incident particle kinetic energy $(E_{\mathrm{K}})_{\mathrm{i}}$, as discussed in Sect 5.3. Energy transfer larger than $\Delta E_{\max}$ is physically impossible.

6.4.2 *Minimum Energy Transfer and Mean Ionization/Excitation Energy*

For large impact parameters b the energy transfer $\Delta E(b)$, calculated from (6.19), may be smaller than the binding energy of the orbital electron or smaller than the minimum excitation energy of the given orbital electron. Thus, no energy transfer is possible for $b > b_{\max}$ where $b_{\max}$ corresponds to a minimum energy transfer $\Delta E_{\min}$, referred to as the mean ionization/excitation energy I of the absorber atom. For a given atom, its mean ionization/excitation energy is always larger than the ionization energy of the atom, since I accounts for all possible atomic ionizations as well as atomic excitations, while the atomic ionization energy (IE), as discussed in Sect. 3.2.4,

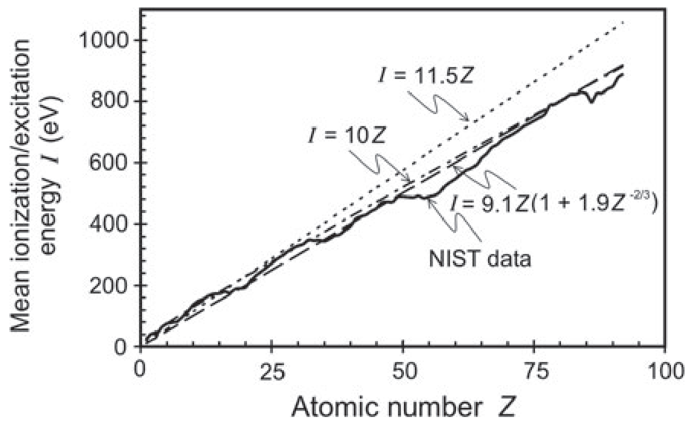


Fig. 6.4 Mean ionization/excitation energy I for elements against atomic number Z. Also plotted are three empirical relationships given in (6.23)–(6.25) that can be used for a rough estimation of I for a given absorber Z. Data were obtained from the ICRU Report 37 and from the NIST

pertains to the energy required to remove the least bound atomic electron (i.e., valence electron in the outer shell).

The mean ionization/excitation energy I corresponds to the minimum amount of energy $\Delta E_{\min}$ that can be transferred, on average, to an absorber atom in a Coulomb interaction between a charged particle and an orbital electron. Using (6.19), $\Delta E_{\min}$ is written as

$$\Delta E_{\min} = 2\left(\frac{e^2}{4\pi\varepsilon_o}\right)^2 \frac{z^2}{m_e \upsilon^2 b_{\max}^2} = I, \tag{6.22}$$

showing that $\Delta E_{\min} \propto 1/b_{\max}^2$ and $b_{\max} \propto 1/\sqrt{\Delta E_{\min}} = 1/\sqrt{I}$.

In general, the mean ionization/excitation energy I cannot be calculated from the atomic theory; however, it can be determined empirically from measured stopping power data compared to data calculated using an appropriate stopping power formula. Current values for I recommended for use by the ICRU and the NIST are plotted in Fig. 6.4 and show a general rise of I with increasing atomic number Z. Figure 6.4 also highlights three empirical relationships that can be used for a rough estimation of I for a given absorber Z. The three approximations are

$$I \text{ (in eV)} = 11.5Z \quad \text{for } Z < 15, \tag{6.23}$$

$$I \text{ (in eV)} = 10Z \quad \text{for } Z > 15, \tag{6.24}$$

$$I \text{ (in eV)} = 9.1Z\left(1 + 1.9Z^{-2/3}\right). \tag{6.25}$$

Typical values of the mean ionization/excitation energy I from the ICRU Report 37 are given in Table 6.3 for several elements and in Table 6.4 for several compounds of interest in medical physics and dosimetry. It is important to note that the mean ionization/excitation energy I only depends on the absorbing medium but does not depend on the type of charged particle interacting with the absorbing medium.

Table 6.3 Mean ionization/excitation energy I for various absorbing materials (from the ICRU Report 37)

Element	H	C	Al	Cu	Ag	W	Pb	Ra	U	Cf
Z	1	6	13	29	47	74	82	88	92	98
I (eV)	19.2	78	167	322	470	727	823	826	890	966

Table 6.4 Mean ionization/excitation energy I for various compounds of interest in medical physics (from the ICRU Report 37)

Compound	I (eV)	Compound	I (eV)
Air (dry)	85.7	Lithium fluoride	94
Water (liquid)	75	Photographic emulsion	331
Water (vapor)	71.6	Sodium iodide	452
Muscle (skeletal)	75.3	Polystyrene	68.7
Bone (compact)	91.9	A-150 plastic	65.1

6.4.3 Maximum Energy Transfer

For small impact parameters b the energy transfer is governed by the maximum energy $\Delta E_{\max}$ that can be transferred in a single head-on collision, as discussed in Sect. 5.3.2. Classically $\Delta E_{\max}$ for a head-on collision between a heavy charged particle M with kinetic energy $E_K = \frac{1}{2}M\upsilon^2$ and an orbital electron with mass m_e where $m_e \ll M$, as discussed in Sect. 5.3.4, is given by

$$\Delta E_{\max} = \frac{4m_e M}{(m_e + M)^2} E_K \approx 4\frac{m_e}{M} E_K = 4\frac{m_e}{M}\frac{M\upsilon^2}{2} = 2m_e\upsilon^2. \tag{6.26}$$

Equation (6.26) shows that only a very small fraction $(4m_e/M)$ of the heavy charged particle kinetic energy can be transferred to an orbital electron in a single collision (note that $M \gg m_e$). The classical relationship between $\Delta E_{\max}$ and the minimum impact parameter $b_{\min}$ that allows the maximum energy transfer from a heavy charged particle to an orbital electron is

$$\Delta E_{\max} = 2\left(\frac{e^2}{4\pi\varepsilon_o}\right)^2 \frac{z^2}{m_e\upsilon^2 b_{\min}^2} = 2m_e\upsilon^2, \tag{6.27}$$

resulting in $\Delta E_{\max} \propto 1/b_{\min}^2$ or $b_{\min} \propto 1/\sqrt{\Delta E_{\max}}$.

In relativistic mechanics the maximum energy transfer $\Delta E_{\max}$ from a heavy charged particle of rest mass M_0 to orbital electron of rest mass m_e where $m_e \ll M_0$ was in (5.41) given as

$$\Delta E_{\max} = 2m_e c^2 \left(\gamma^2 - 1\right) \frac{1}{1 + 2\gamma \dfrac{m_e}{M_0} + \left(\dfrac{m_e}{M_0}\right)^2} \approx 2m_e c^2 \frac{\beta^2}{1-\beta^2}, \tag{6.28}$$

where β and γ are given by the standard relativistic relationships

$$\beta = \frac{\upsilon}{c}; \quad \gamma = \frac{1}{\sqrt{1-\beta^2}}; \quad \text{and} \quad \gamma^2 - 1 = \frac{\beta^2}{1-\beta^2}. \tag{6.29}$$

6.4.4 Classical Derivation of Mass Collision Stopping Power

Following the general definition of the mass collision stopping power given in (6.21), integration over the impact parameter b or energy transfer $\Delta E(b)$ gives, respectively

$$S_{col} = -\frac{1}{\rho}\frac{dE}{dx} = N_e \int \Delta E(b) d\sigma_{col} = 2\pi N_e \int_{b_{\min}}^{b_{\max}} \Delta E(b) b db \tag{6.30}$$

and

$$\begin{aligned} S_{col} &= -\frac{1}{\rho}\frac{dE}{dx} = N_e \int \Delta E(b)\, d\sigma_{col} \\ &= 2\pi N_e \int_{\Delta E_{\min}}^{\Delta E_{\max}} \Delta E(b) \frac{b\, db}{d[\Delta E(b)]} d[\Delta E(b)], \end{aligned} \tag{6.31}$$

where we integrate (6.30) from the minimum impact parameter $b_{\min}$ to maximum impact parameter $b_{\max}$ and (6.31) from minimum energy transfer $\Delta E_{\min}$ to maximum energy transfer $\Delta E_{\max}$, and

N_e is the electron density, i.e., number of electrons per mass of the absorber ($N_e = ZN_A/A$), with Z and A the atomic number and atomic mass number of the absorber, respectively.

$d\sigma_{col}$ is the differential cross section for collision loss expressed as

$$d\sigma_{col} = 2\pi b\, db = 2\pi b \frac{db}{d[\Delta E(b)]} d[\Delta E(b)], \tag{6.32}$$

with b given in (6.20) and the derivative $db/d[\Delta E(b)]$ equal to

$$\frac{db}{d[\Delta E(b)]} = \frac{1}{2}\sqrt{\frac{2}{m_e}} \left(\frac{e^2}{4\pi\varepsilon_0}\right) \frac{z}{\upsilon} [\Delta E(b)]^{-3/2}, \tag{6.33}$$

to give the following expression for $\mathrm{d}\sigma_{\mathrm{col}}$

$$\mathrm{d}\sigma_{\mathrm{col}} = 2\pi b\,\mathrm{d}b = 2\pi b \frac{\mathrm{d}b}{\mathrm{d}[\Delta E(b)]}\mathrm{d}[\Delta E(b)]$$
$$= 2\pi \left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\mathrm{e}}\upsilon^2} \frac{\mathrm{d}[\Delta E(b)]}{[\Delta E(b)]^2}. \tag{6.34}$$

After incorporating (6.19) into (6.30), the mass collision stopping power S_{col} can be written as

$$S_{\mathrm{col}} = 4\pi \frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\mathrm{e}}\upsilon^2}\int_{b_{\min}}^{b_{\max}} \frac{\mathrm{d}b}{b} = 4\pi \frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\mathrm{e}}\upsilon^2}\ln\frac{b_{\max}}{b_{\min}}, \tag{6.35}$$

while, after incorporating (6.34) into (6.31), we get the following expression for the mass collision stopping power S_{col}

$$S_{\mathrm{col}} = 2\pi \frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\mathrm{e}}\upsilon^2}\int_{\Delta E_{\min}}^{\Delta E_{\max}} \frac{\mathrm{d}[\Delta E(b)]}{[\Delta E(b)]}$$
$$= 2\pi \frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\mathrm{e}}\upsilon^2}\ln\frac{\Delta E_{\max}}{\Delta E_{\min}}. \tag{6.36}$$

Since from (6.22) and (6.27) we write

$$\frac{b_{\max}}{b_{\min}} = \sqrt{\frac{\Delta E_{\max}}{\Delta E_{\min}}}, \tag{6.37}$$

and obtain the following expression for S_{col} of (6.35)

$$S_{\mathrm{col}} = 4\pi \frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\mathrm{e}}\upsilon^2}\ln\frac{b_{\max}}{b_{\min}}$$
$$= 4\pi \frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\mathrm{e}}\upsilon^2}\ln\sqrt{\frac{\Delta E_{\max}}{\Delta E_{\min}}}, \tag{6.38}$$

identical to S_{col} in (6.36).

The energy transfer $\Delta E(b)$ from the heavy charged particle to an orbital electron ranges from $\Delta E_{\min}\,(b_{\max}) = I$ to $\Delta E_{\max}\,(b_{\min})$ of (6.27). Insertion of expressions for $\Delta E_{\min}$ and $\Delta E_{\max}$ given in (6.22) and (6.27), respectively, into (6.36) or (6.38) results in the following classical expression for mass collision stopping power derived by *Niels Bohr* in 1913

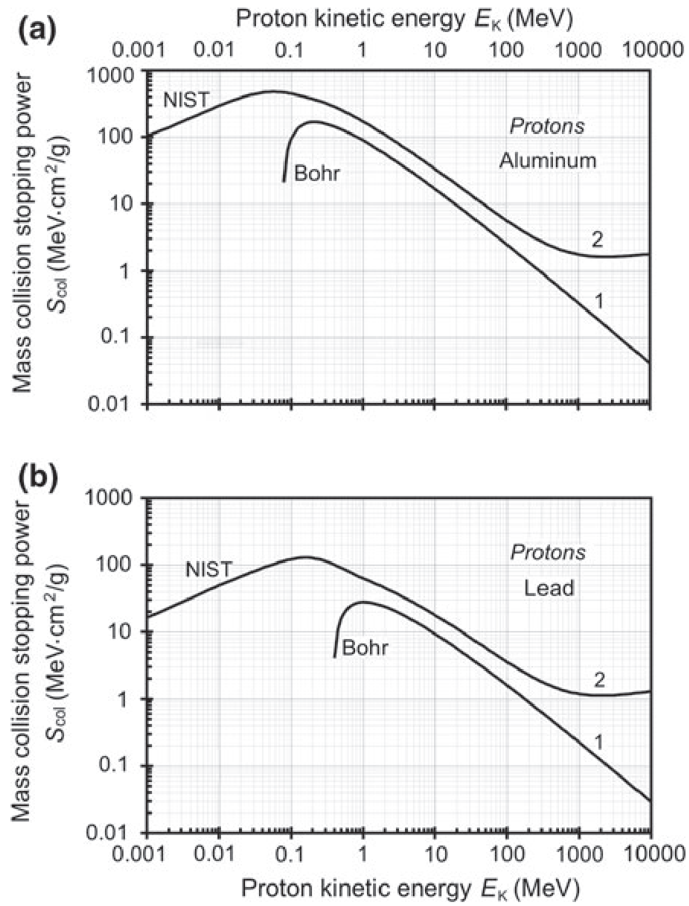


Fig. 6.5 Mass collision stopping powers of *part* **a** aluminum and *part* **b** lead against proton kinetic energy E_K. Curves (*2*) represent experimental data compiled by the NIST, curves (*1*) are calculated with Bohr's stopping power equation (6.39)

$$S_{col} = 2\pi \frac{ZN_A}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_e\upsilon^2} \ln \frac{2m_e\upsilon^2}{I}. \tag{6.39}$$

Figure 6.5 shows a plot of the mass collision stopping power S_{col} of aluminum in part (a) and lead in part (b) for protons ($z = 1$) in the kinetic energy range from 10^{-3} MeV to 10^4 MeV. Curves (2) are for measured data tabulated by the NIST, curves (1) are for the Bohr classical expression given in (6.39).

In the intermediate energy range from ∼300 keV to ∼100 MeV both the theory and measurement exhibit the same trends; however, the measurement consistently exceeds the calculated data by a factor of 2. In the low energy range ($E_K < 300$ keV) and the high energy range ($E_K > 100$ MeV) the discrepancy between measured and calculated data is significantly larger, clearly indicating that Bohr classical theory does not provide a realistic description of particle stopping in absorbing media. The reasons for this seem obvious: the Bohr theory ignores quantum mechanical and relativistic effects and treats electron binding effects in a very rudimentary fashion through the mean ionization/excitation energy I.

6.4.5 Bethe Collision Stopping Power

With the advent of quantum mechanics and relativistic physics many scientists attempted to improve the Bohr collision stopping power theory by incorporating the new concepts into the newly developed theories. Most notable were efforts in 1930s by *Hans Bethe* and *Felix Bloch* who developed a new collision stopping power theory based on quantum mechanical and relativistic concepts and, in contrast to Bohr classical theory, achieved excellent agreement between theoretical and experimental data.

In 1931 *Hans Bethe* proposed a theory in which the energy loss of a heavy charged particle traversing an absorber is calculated using the Born approximation applied to the collision of the heavy charged particle with atomic orbital electrons. As suggested in Sect. 1.23.6, the differential cross section for momentum transfer from the heavy charged particle to atomic electrons was proportional to the square of the matrix element defined by the Coulomb interaction between the relevant initial and final states, and the wave functions for the incident and scattered heavy charged particle were described by plane waves.

Bethe divided the possible energy transfers ΔE to an orbital electron into two categories: soft (distant) and hard (close). The energy boundary η between soft and hard collisions was not clearly defined but was generally chosen such that for soft collisions the energy transfer ΔE was smaller than η and for hard collisions ΔE was larger than η. Thus, η was large compared to the binding energies of atomic electrons, yet it was also small enough to allow for all hard collisions, characterized with $\Delta E \geq \eta$, a representation of the charged particle as point source.

Bethe considered average energy losses for the soft and hard collisions separately and obtained the following expressions for the soft and hard mass collision stopping powers

$$S_{\mathrm{col}}^{\mathrm{soft}} = \left(-\frac{1}{\rho}\frac{\mathrm{d}E}{\mathrm{d}x}\right)_{\Delta E<\eta}^{\mathrm{soft}} = 2\pi\frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2\frac{z^2}{m_{\mathrm{e}}\beta^2c^2}\left\{\ln\frac{2m_{\mathrm{e}}c^2\beta^2\eta}{\left(1-\beta^2\right)I^2}-\beta^2\right\} \tag{6.40}$$

and

$$S_{\mathrm{col}}^{\mathrm{hard}} = -\frac{1}{\rho}\left(\frac{\mathrm{d}E}{\mathrm{d}x}\right)_{\Delta E>\eta}^{\mathrm{hard}} = 2\pi\frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2\frac{z^2}{m_{\mathrm{e}}\beta^2c^2}\left\{\ln\frac{\Delta E_{\mathrm{max}}}{\eta}-\beta^2\right\}$$

$$= 2\pi\frac{ZN_{\mathrm{A}}}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2\frac{z^2}{m_e\beta^2c^2}\left\{\ln\frac{2m_{\mathrm{e}}c^2\beta^2}{\left(1-\beta^2\right)\eta}-\beta^2\right\}, \tag{6.41}$$

where

I is the mean ionization/excitation energy, discussed in Sect. 6.4.2.
ΔE_{max} is the maximum energy transfer from the heavy charged particle to the orbital electron, as given relativistically in (5.4.1) and (6.28).

The expression for $S_{\text{col}}^{\text{soft}}$ given in (6.40) is valid for all types of charged particles (heavy and light), while the expression for $S_{\text{col}}^{\text{hard}}$ given in (6.41) is valid only for heavy charged particles and cannot be used for electrons and positrons.

As suggested in (6.3), the mass collision stopping power is given as the sum of the soft (6.40) and hard (6.41) mass collision stopping powers. Adding (6.40) and (6.41) we get

$$\begin{aligned} S_{\text{col}} = S_{\text{col}}^{\text{soft}} + S_{\text{col}}^{\text{hard}} &= 2\pi \frac{ZN_{\text{A}}}{A} \left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\text{e}}c^2\beta^2} \\ &\times \left\{ \ln \frac{2m_{\text{e}}c^2\beta^2\eta}{\left(1-\beta^2\right) I^2} + \ln \frac{2m_{\text{e}}c^2\beta^2}{\left(1-\beta^2\right)\eta} - 2\beta^2 \right\} \\ &= 4\pi \frac{ZN_{\text{A}}}{A} \left(\frac{c^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\text{e}}e^2\beta^2} \left\{ \ln \frac{2m_e c^2}{I} + \ln \frac{\beta^2}{1-\beta^2} - \beta^2 \right\} \\ &= C_0 \frac{z^2}{A\beta^2} Z \left\{ \ln \frac{2m_{\text{e}}c^2}{I} + \ln \frac{\beta^2}{1-\beta^2} - \beta^2 \right\}, \end{aligned} \tag{6.42}$$

where the demarcation energy η separating soft from hard collisions falls out of the final equation and C_0 is the collision stopping power constant equal to

$$C_0 = 4\pi N_{\text{A}} \left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{1}{m_{\text{e}}c^2} = 4\pi N_{\text{A}} r_{\text{e}}^2 m_{\text{e}} c^2 = 0.3071\ \text{MeV}\cdot\text{cm}^2\cdot\text{mol}^{-1}, \tag{6.43}$$

with $r_{\text{e}} = e^2/\left(4\pi\varepsilon_0 m_{\text{e}} c^2\right) = 2.818$ fm defined as the classical electron radius.

The expression (6.42) is known as the Bethe mass collision stopping power equation. It is valid for heavy charged particles, such as protons and α-particles, and accounts for quantum mechanical as well as relativistic effects. In the classical limit for $\upsilon \ll c$ (6.42) yields

$$S_{\text{col}} = 4\pi \frac{ZN_{\text{A}}}{A} \left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{z^2}{m_{\text{e}}\upsilon^2} \ln \frac{2m\upsilon^2}{I} = C_0 \frac{z^2c^2}{A\upsilon^2} Z \ln \frac{2m_{\text{e}}\upsilon^2}{I}, \tag{6.44}$$

that is non-relativistic yet it accounts for quantum-mechanical effects and is exactly double the result obtained from the Bohr classical stopping power theory in (6.39). Since, as shown in Fig. 6.5, the discrepancy between Bohr theory and measurement in the intermediate energy range was also by a factor of 2, one may surmise that a plot of mass collision stopping power calculated from the Bethe equation (6.42) will yield good agreement with measured data. That this is so is shown in Fig. 6.6 in which we plot the measured (available from the NIST) and calculated mass collision stopping powers S_{col} of aluminum for protons in the kinetic energy range from 1 keV to 10^4 MeV. The calculated data are obtained from two different theories: Bohr classical theory data are calculated from (6.39) and shown by the dashed curve and Bethe theory data are calculated from (6.42) and shown by the solid curve.

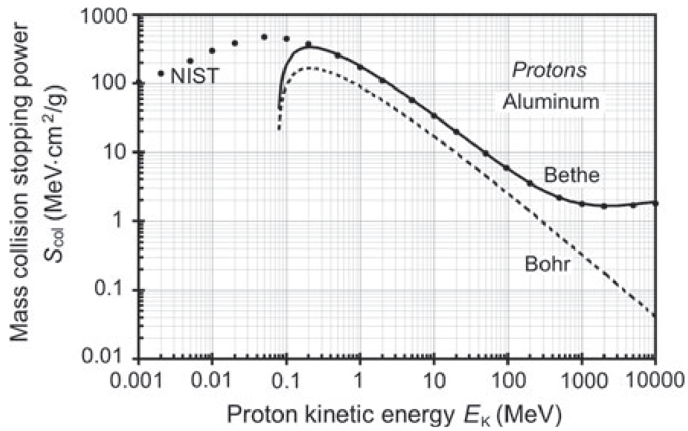


Fig. 6.6 Measured and calculated mass collision stopping powers of aluminium against proton kinetic energy E_K. Experimental data available from the NIST are shown with data points; calculated data are shown with *solid* and *dashed curves*. The *dashed curve* is calculated with the Bohr classical mass collision stopping power equation (6.39); the *solid curve* is calculated with the Bethe mass collision stopping power equation (6.42) accounting for quantum mechanical as well as relativistic effects

As shown in Fig. 6.6, in the intermediate and high energy relativistic range the agreement between experiment and Bethe theory is excellent; however, at low energies the agreement fails. Both the theory and measurement predict a peak in stopping power with decreasing energy, but the measured peak is higher and appears at a lower kinetic energy than the calculated peak. Clearly there must be some other unaccounted for effect that causes this discrepancy. As discussed in Sect. 6.4.6, two corrections (shell correction and density effect correction) are applied to Bethe theory in order to explain this discrepancy at low kinetic energies.

The discrepancy at low kinetic energies notwithstanding, the Bethe collision stopping power formula agrees well with measured data in the intermediate and high energy relativistic region, and clearly predicts the shape of the stopping power curve as well as provides the functional dependence of the many important parameters of the charged particles and absorbing media that influence the stopping power curve.

Shape of the Bethe Mass Collision Stopping Power Curve

As indicated schematically in Fig. 6.7, a plot of the collision stopping power S_{col} against kinetic energy E_K for a heavy charged particle goes through three distinct regions with increasing E_K:

Region 1: At very low kinetic energies, S_{col} rises with energy and reaches a peak.

Region 2: In the intermediate energy region beyond the peak, S_{col} decreases as $1/\upsilon^2$ or $1/E_K$ of the charged particle until it reaches a broad minimum.

Region 3: In the relativistic energy region beyond the broad minimum, S_{col} rises slowly with increasing kinetic energy E_K as a result of the relativistic terms $\left\{\ln\beta^2 - \ln(1-\beta^2) - \beta^2\right\}$.

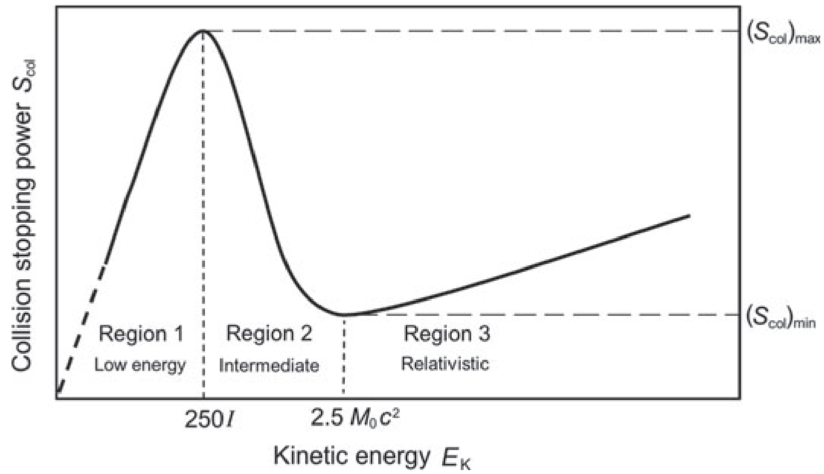


Fig. 6.7 Schematic representation of the shape of the collision stopping power S_{col} as a function of the charged particle kinetic energy E_K. Three regions are shown as the kinetic energy increases from zero: (**1**) low energy region; (**2**) intermediate energy region; and (**3**) relativistic energy region. In region (**1**), S_{col} rises almost linearly and reaches a maximum at about 250 I, where I is the mean ionization/excitation energy of the absorber. In region (**2**), S_{col} decreases as $1/\upsilon^2$ or $1/E_K$ where υ is the velocity of the charged particle to reach a broad minimum at $\sim 2.5 M_0c^2$ where M_0c^2 is the rest energy of the charged particle. In region (**3**), S_{col} rises slowly with increasing E_K because of relativistic effects

Maxima and Minima on the Stopping Power Curves

The maximum $S_{col}^{max}(E_K)$ and minimum $S_{col}^{min}(E_K)$ on the stopping power curve (6.42) can be determined by setting $\mathrm{d}S_{col}/\mathrm{d}E_K = 0$. To simplify the calculation we note that $\mathrm{d}S_{col}/\mathrm{d}E_K = (\mathrm{d}S_{col}/\mathrm{d}\beta)\,(\mathrm{d}\beta/\mathrm{d}E_K)$ and $\mathrm{d}\beta/\mathrm{d}E_K \neq 0$. We then write the Bethe equation (6.42) as

$$S_{col} = \frac{\text{const}}{\beta^2}\left\{\ln\frac{2m_ec^2}{I} + \ln\beta^2 - \ln(1-\beta^2) - \beta^2\right\} \tag{6.45}$$

and determine $\mathrm{d}S_{col}/\mathrm{d}\beta$ as

$$\begin{aligned}\frac{\mathrm{d}S_{col}}{\mathrm{d}\beta} &= -\frac{2\,\text{const}}{\beta^3}\left\{\ln\frac{2m_ec^2}{I} + \ln\beta^2 - \ln(1-\beta^2) - \beta^2\right\}\\ &\quad + \frac{\text{const}}{\beta^2}\left\{\frac{2}{\beta} + \frac{2\beta}{1-\beta^2} - 2\beta\right\} = 0\end{aligned} \tag{6.46}$$

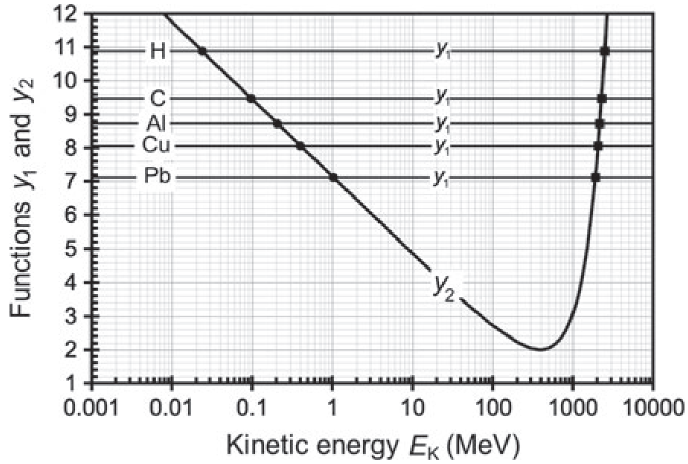


Fig. 6.8 Functions y_1 and y_2 against kinetic energy E_K of the heavy charged particle. Function y_1 is a constant for a given element and is shown for various elements between hydrogen ($Z = 1$) and lead ($Z = 82$). The left intercepts between the two functions indicate the maximum in stopping power according to Bethe collision stopping power equation (6.42), the right intercepts indicate the minimum

or

$$\ln\frac{2m_ec^2}{I} = 1 + \frac{\beta^2}{1-\beta^2} - \ln\frac{\beta^2}{1-\beta^2} = \frac{1}{1-\beta^2} - \ln\frac{\beta^2}{1-\beta^2}$$
$$= 1 + \frac{E_K}{E_0}\left(2 + \frac{E_K}{E_0}\right) - \ln\left\{\frac{E_K}{E_0}\left(2+\frac{E_K}{E_0}\right)\right\}, \tag{6.47}$$

where we used $E_K/E_0 = 1/\sqrt{1-\beta^2}$ with E_K and E_0 kinetic energy and rest energy, respectively, of the incident heavy charged particle of rest mass M_0.

We now have two functions, y_1 and y_2, where

$$y_1 = \ln\frac{2m_ec^2}{I} \tag{6.48}$$

depends on the mean ionization/excitation energy I and

$$y_2 = \frac{1}{1-\beta^2} - \ln\frac{\beta^2}{1-\beta^2} = 1 + \frac{E_K}{E_0}\left(2 + \frac{E_K}{E_0}\right) - \ln\left\{\frac{E_K}{E_0}\left(2+\frac{E_K}{E_0}\right)\right\} \tag{6.49}$$

depends on the charged particle kinetic energy E_K and rest energy E_0.

A plot of the two functions, shown in Fig. 6.8 for a full range of absorbers from $Z = 1$ to $Z = 92$, for a given absorber results in two intercepts between the two functions; intercept on the left defines E_K for $S_{col}^{max}(E_K)$ and the one on the right defines E_K for $S_{col}^{min}(E_K)$. While the region of E_K for $S_{col}^{max}(E_K)$ seems relatively broad, ranging from 0.02 MeV for hydrogen to 1 MeV for lead, the region of E_K for

$S_{\text{col}}^{\min}(E_{\text{K}})$ is very narrow ranging from $\sim 2\times 10^3$ MeV for uranium to $\sim 2.4\times 10^3$ MeV for hydrogen. The minimum in the collision stopping power curve is thus essentially independent of absorber material and occurs at $E_{\text{K}} \approx 2.5E_0 = 2.5\, M_0c^2$.

$S_{\text{col}}^{\max}(E_{\text{K}})$ is in the low kinetic energy region where $E_{\text{K}} \approx M_0c^2$ and can be evaluated by assuming that $E_{\text{K}} = \frac{1}{2}M_0c^2\beta^2$ or $\beta^2 = 2E_{\text{k}}/(M_0c^2)$. This results in significantly simpler form of function y_2 that can be approximated as $y_2 \approx 1 - \ln\beta^2 \approx 1 - \ln\{2E_{\text{k}}/(M_0c^2)\}$. The (y_1, y_2) intercept that signifies $S_{\text{col}}^{\max}$ is then determined as

$$\ln\frac{2m_{\text{e}}c^2}{I} = y_1 \equiv y_2 \approx 1 - \ln\frac{2E_{\text{K}}}{M_0c^2} \tag{6.50}$$

or

$$E_{\text{K}} = \frac{eM_0c^2}{4m_{\text{e}}c^2}I \approx 1250\, AI, \tag{6.51}$$

where we use $M_0 \approx Am_{\text{p}}$. For the mass of the heavy charged particle, with m_{p} the proton mass and A the atomic mass number of the heavy charged particle; $m_{\text{p}}/m_{\text{e}} = 1836$; and $e = 2.718$ the base of natural logarithm.

Figure 6.5 shows the proton beam maxima for aluminum and lead at 207.5 keV and 1.03 MeV, respectively, significantly higher than experimental data which show the maxima at 50 keV and 200 keV, respectively. An inspection of stopping power tables for various absorber elements shows that the experimentally measured E_{K} for $S_{\text{col}}^{\max}(E_{\text{K}})$ increases with atomic number Z of the absorber and occurs at approximately $250I$ for all absorber elements.

Dependence of Collision Stopping Power on Particle Charge

S_{col} of (6.42) depends on z^2, the atomic number of the heavy charged particle (projectile). This implies, for example, that mass collision stopping powers of an absorbing medium for deuteron and proton of same velocity β will be the same, i.e., $S_{\text{d}}(\beta) = S_{\text{p}}(\beta)$ while for α-particle and proton of same velocity β they will differ by a factor of 4, i.e., $S_\alpha(\beta) = 4S_{\text{p}}(\beta)$.

Dependence of Collision Stopping Power on Particle Velocity

At low kinetic energies the Bethe equation (6.42) breaks down, as shown in Fig. 6.6, and to achieve better agreement between theory and measurement one must use several correction factors, discussed in Sect. 6.4.6. At intermediate energies S_{col} is governed by the $1/\upsilon^2$ (i.e., $1/E_{\text{K}}$) term and decreases rapidly with increasing E_{K}. In the relativistic energy region S_{col} rises slowly with kinetic energy as a result of the $1/\beta^2$ term which saturates at ~ 1 and the $\left\{\ln[\beta^2/(1-\beta^2)] - \beta^2\right\}$ term that slowly increases with increasing kinetic energy.

Dependence of Collision Stopping Power on Particle Mass

As shown in (6.42), S_{col} does **not** depend on the mass of the heavy charged particle. A given absorbing material will have the same collision stopping power for all heavy charged particles of a given velocity and charge ze.

Dependence of Collision Stopping Power on Absorbing Medium

S_{col} of (6.42) depends on the atomic number Z, atomic mass A, and mean ionization/excitation energy I of the absorbing medium through two terms, both of which decrease the mass collision stopping power with an increase in atomic number Z. The first term is the Z/A term of the electron density $N_e = ZN_A/A$ and the second term is the $(-\ln I)$ term.

Since Z/A varies from substance to substance within quite a narrow range (it falls from 0.5 for low Z elements to $\sim$0.4 for high Z elements, with one notable exception of hydrogen for which $Z/A \approx 1$), we note that the mass collision stopping power does not vary much from substance to substance. This means that the energy losses of a given charged particle passing through layers of equal thickness in g/cm^2 are about the same for all substances. One should note, however, that, for a given charged particle kinetic energy, S_{col} of lower atomic number absorber will exceed S_{col} of a higher atomic number absorber.

6.4.6 Corrections to Bethe Collision Stopping Power Equation

As discussed in relation to Fig. 6.6, the Bethe collision equation is not in complete agreement with experimental data and various approximations have been developed to correct the problem, most notably:

(i) the so-called shell correction applicable at relatively low charged particle kinetic energy where the particle velocity is comparable to the velocity of orbital electrons of the absorber atoms.
(ii) Barkas-Bloch correction accounting for departures from the first order Born approximation.
(iii) density effect correction that accounts for the polarization of the absorber medium and is important at high kinetic energies of the charged particle traversing the absorber.

Shell Correction

Bethe's approach, based on the Born approximation, assumes that the velocity of the heavy charged particle is much larger than the velocity of the bound orbital electrons of the absorber. While at high kinectic energies this assumption is correct, at low kinetic energies of the incident charged particle it does not hold. Orbital electrons stop participating in energy transfer from the charged particle when their velocity becomes comparable to the charged particle velocity. This effect causes an overestimate in the mean ionization/excitation energy I at low energies and results in an underestimate in S_{col} calculated from the Bethe equation.

The so-called shell correction term C/Z is introduced to account for the overestimate in I and various theories have subsequently been developed for its determination. Since the K shell electrons are the fastest of all orbital electrons, they are the first to be affected by low particle velocity with decreasing particle velocity. The shell correction is often addressed as the K shell correction, labeled C_K/Z and all possible higher shell corrections are usually ignored.

The correction term C/Z is a function of the absorbing medium and charged particle velocity; however, for the same medium and particle velocity, it is the same for all particles including electrons and positrons.

Polarization or Density Effect Correction

In addition to the shell correction factor, a second correction term δ is applied to the Bethe collision stopping power equation to account for the polarization or density effect in condensed media. The effect influences the soft (distant) collision interactions by polarizing the condensed absorbing medium thereby decreasing the collision stopping power of the condensed medium in comparison with same absorbing medium in gaseous state. For heavy charged particles the density correction is important at relativistic energies and negligible at intermediate and low energies; however, for electrons and positrons it plays a role in stopping power formulas at all energies.

Bethe Collision Stopping Power Equation Incorporating Shell Correction and Density Effect Correction

The Bethe collision stopping power equation incorporating the shell correction and density correction terms is for heavy charged particles written as

$$S_{\mathrm{col}} = 4\pi \frac{N_A}{A} \left(\frac{e^2}{4\pi\varepsilon_0} \right)^2 \frac{z^2}{m_e c^2 \beta^2} Z \left\{ \ln \frac{2m_e c^2}{I} + \ln \frac{\beta^2}{1-\beta^2} - \beta^2 - \frac{C}{Z} - \delta \right\}, \tag{6.52}$$

where C/Z and δ are the shell correction and density effect correction, respectively, and all other parameters of the absorber (Z, A, and I) and of the heavy charged particle (β and z) were defined before in (6.42).

6.4.7 Collision Stopping Power Equations for Heavy Charged Particles

The collision stopping power equations stated above for heavy charged particles can be summarized as follows

$$S_{\mathrm{col}} = C_0 \frac{z^2}{\beta^2} \frac{Z}{A} B_{\mathrm{col}}, \tag{6.53}$$

where

C_0 is the collision stopping power constant defined in (6.43) with the following expression $C_0 = 4\pi N_A r_e^2 m_e c^2 = 0.3071\,\mathrm{MeV{\cdot}cm^2{\cdot}mol^{-1}}$.
z is the atomic number of the heavy charged particle.
β is the heavy charged particle velocity normalized to the speed of light c.

Table 6.5 Expressions for the atomic stopping number B_{col} for various energy ranges of heavy charged particle energy

Derivation of S_{col}	Atomic stopping number B_{col}	
Classical Bohr – 1913	$\left\{\ln\sqrt{\frac{2m_e\upsilon^2}{I}}\right\}$	(6.54)
Non-relativistic, quantum-mechanical Bethe – 1931	$\left\{\ln\frac{2m_e\upsilon^2}{I}\right\}$	(6.55)
Relativistic, quantum-mechanical Bethe – 1931	$\left\{\ln\frac{2m_ec^2}{I}+\ln\frac{\beta^2}{1-\beta^2}-\beta^2\right\}$	(6.56)
Relativistic, quantum-mechanical with shell correction (C/Z) and density effect correction (δ)	$\left\{\ln\frac{2m_ec^2}{I}+\ln\frac{\beta^2}{1-\beta^2}-\beta^2-\frac{C}{Z}-\delta\right\}$	(6.57)

Z is the atomic number of the absorbing medium.
A is the atomic mass of the absorber in g/mol.
B_{col} is the so-called *atomic stopping number*.

The atomic stopping number B_{col} is a function of the velocity β of the charged particle and of the atomic number Z of the absorber through the mean ionization/excitation energy I. The form of the expression for B_{col} also depends on the specific approach taken in its derivation, as indicated in Table 6.5.

The units of S_{col} in (6.53) are MeV·cm^2·g^{-1}; the constant C_0 has units of MeV·cm^2·mol^{-1}, and B_{col} has no units. Since the units of A are g/mol, incorporating an appropriate value for A into (6.53) results in proper units for S_{col} in MeV·cm^2·g^{-1}. From the general expression for the mass collision stopping power S_{col} given in (6.53) it is evident that S_{col} for a heavy charged particle traversing an absorber *does not depend on charged particle mass* but depends upon charged particle velocity β and atomic number z as well as absorber atomic number Z, atomic mass A, and mean ionization/excitation energy I.

Figure 6.9 plots the mass collision stopping powers of carbon, aluminum, copper, and lead for protons in the kinetic energy E_K range from 1 keV to 10^4 MeV. The data were obtained from the NIST which at high proton energies uses Bethe collision stopping power equation (6.42) including shell and density correction terms as well as the so-called Barkas and Bloch corrections for deviations from the Born approximation. At low proton energies the NIST data are determined with fitting formulas which are largely based on experimental data. The NIST collision stopping power database uses mean ionization/excitation energies recommended by the ICRU Report 37.

The four curves in Fig. 6.9 exhibit the standard collision stopping power behavior of rising with E_K at low kinetic energies, reaching a peak at between 50 keV and

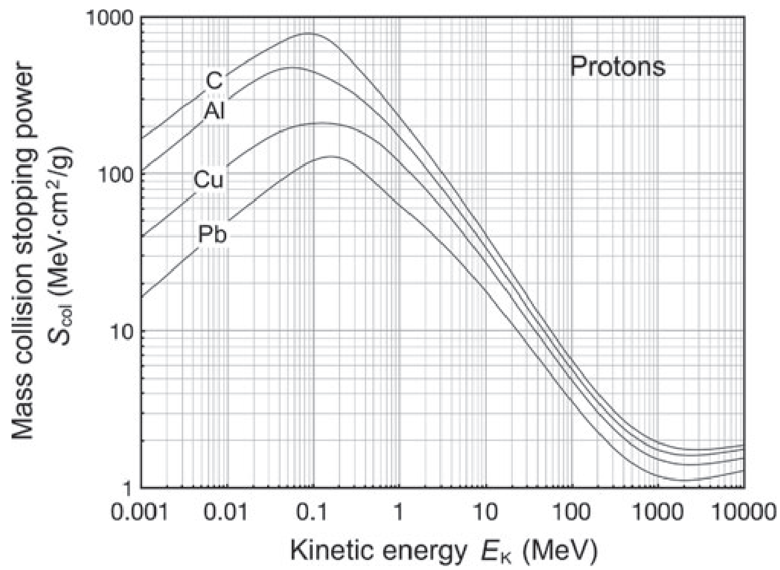


Fig. 6.9 Mass collision stopping powers of carbon, aluminum, copper, and lead against proton beam kinetic energy. Data were obtained from the NIST

200 keV, then falling as $1/\upsilon^2$ or $1/E_K$ at intermediate energies, reaching a minimum around $E_K/E_0 \approx 2.5$, and then rising slowly with E_K in the relativistic energy range. In the intermediate and relativistic energy range the mass collision stopping powers at a given kinetic energy E_K are fairly similar; however, it is evident that lower atomic number absorbers have higher mass collision stopping powers because of the effects of the (Z/A) and $(-\ln I)$ terms in the Bethe equation (6.42), as discussed in Sect. 6.4.6.

6.5 Collision Stopping Power for Light Charged Particles

Electron and positron interactions (collisions) with orbital electrons of an absorber differ from those of heavy charged particles in three important aspects. For light charged particles:

1. Relativistic effects become important at relatively low kinetic energies.
2. Collisions with orbital electrons may result in large energy transfers (up to 50% of the incident energy for electrons, up to 100% of the incident energy for positrons). Collisions may also result in elastic and inelastic scattering with large angular deviations.
3. Collisions with nuclei of the absorber may result in bremsstrahlung production (radiation loss, see Sect. 6.3) and, depending on the light charged particle incident energy, radiation loss may actually exceed the collision loss.

In a manner similar to the approach taken with heavy charged particles, S_{col} for electrons and positrons is also calculated separately for soft and hard collisions, and the results are then added to obtain the collision stopping power S_{col}, as indicated in (6.3). The soft collision term ($\Delta E < \eta$) is identical to that obtained for heavy charged particles in (6.40) and the integration also runs from I to η except that for electrons and positrons we use $z = 1$ to obtain

$$S_{col}^{soft} = 2\pi \frac{ZN_A}{A}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{1}{m_e c^2\beta^2}\left\{\ln \frac{2m_e c^2\beta^2\eta}{(1-\beta^2)I^2} - \beta^2\right\}. \tag{6.58}$$

On the other hand, the hard collision term ($\Delta E > \eta$) is significantly more complicated for light charged particles than the term given for heavy charged particles in (6.41). The hard collision term for light charged particles is calculated by integrating

$$S_{col}^{hard} = N_e \int_{\eta}^{\Delta E_{max}} \Delta E \, d\sigma_{col} = N_e \int_{\eta}^{\Delta E_{max}} \Delta E \frac{d\sigma_{col}}{d(\Delta E)} d(\Delta E), \tag{6.59}$$

with the differential cross section for electrons determined by *Christian Møller* and for positrons by *Homi J. Bhabha*. The integration runs from η to ΔE_{max} where $\Delta E_{max} = \frac{1}{2}E_K$ for electrons and $\Delta E_{max} = E_K$ for positrons, as discussed in Sect. 5.3.4.

The complete mass collision stopping powers for electrons and positrons, according to the ICRU Report 37, are expressed as follows

$$S_{col} = 2\pi r_e^2 \frac{Z}{A} N_A \frac{m_e c^2}{\beta^2}\left\{\ln \frac{E_K^2}{I^2} + \ln\left(1 + \frac{\tau}{2}\right) + F^{\pm}(\tau) - \delta\right\}, \tag{6.60}$$

where r_e is the classical electron radius defined in (6.43). In (6.60) the function $F^-(\tau)$ applies to electrons and is given as

$$F^-(\tau) = \left(1-\beta^2\right)\left[1 + \tau^2/8 - (2\tau+1)\ln 2\right], \tag{6.61}$$

while the function $F^+(\tau)$ applies to positrons and is given as

$$F^+(\tau) = 2\ln 2 - \left(\beta^2/12\right)\left[23 + 14/(\tau+2) + 10/(\tau+2)^2 + 4/(\tau+2)^3\right], \tag{6.62}$$

where

τ is the electron or positron kinetic energy E_K normalized to $m_e c^2$, i.e., $\tau = E_K/(m_e c^2)$, as discussed in Sect. 2.7.4.

β is the electron or positron velocity normalized to c, i.e., $\beta = \upsilon/c$.

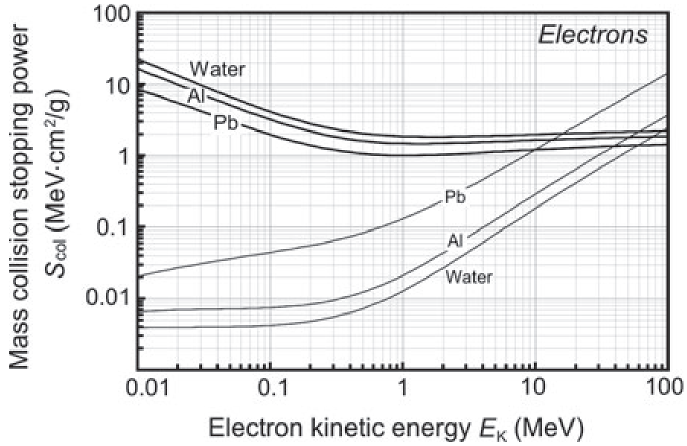


Fig. 6.10 Mass collision stopping power S_{col} for electrons in water, aluminum and lead against electron kinetic energy. The collision stopping power data are shown with heavy solid curves; the radiation stopping power data of Fig. 6.2 are shown with light solid curves for comparison. Data were obtained from the NIST

Figure 6.10 shows mass collision stopping powers S_{col} for electrons in water, aluminum and lead with heavy solid lines and, for comparison, mass radiation stopping powers of Fig. 6.2 are shown with light solid lines. Similarly to stopping power behavior for heavy charged particles, the data of Fig. 6.10 show that higher atomic number absorbers have lower S_{col} than lower atomic number absorbers at same electron energies. The dependence of S_{col} on stopping medium results from two factors in the stopping power expression given by (6.60), both lowering S_{col} with an increasing Z of the stopping medium:

1. The factor Z/A makes S_{col} dependent on the number of electrons per unit mass of the absorber. Z/A is 1 for hydrogen; 0.5 for low Z absorbers; then gradually drops to $\sim$0.4 for high Z absorbers.
2. The $(-\ln I)$ term decreases S_{col} with increasing Z, since I increases almost linearly with increasing Z, as shown in (6.23)–(6.25).

6.6 Total Mass Stopping Power

Generally, the total mass stopping power S_{tot} of an absorber is given by the sum of two components: the mass radiation stopping power S_{rad} and the mass collision stopping power S_{col}

$$S_{tot} = S_{rad} + S_{col}. \tag{6.63}$$

For heavy charged particles the radiation stopping power is negligible ($S_{rad} \approx 0$), thus $S_{tot} = S_{col}$.

For light charged particles both components contribute to the total stopping power; within a broad range of kinetic energies below 10 MeV collision (ionization) losses are dominant ($S_{col} > S_{rad}$); however, the situation is reversed at high kinetic energies where $S_{rad} > S_{col}$. The following features are noted:

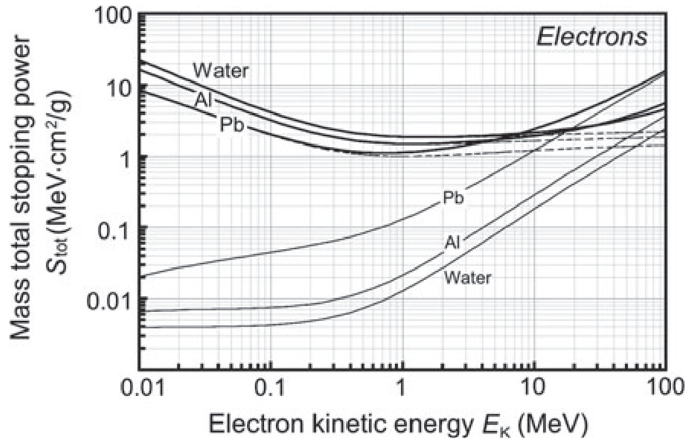


Fig. 6.11 Total mass stopping power S_{tot} for electrons in water, aluminum and lead against the electron kinetic energy shown with heavy solid curves. The mass collision stopping powers and mass radiation stopping powers are shown with dashed curves and light solid curves, respectively, for comparison. The total stopping power of a given material is the sum of the radiation and collision stopping powers. Data were obtained from the NIST. The critical energies $(E_{\text{K}})_{\text{crit}}$ are 100 MeV, 61 MeV, and 10 MeV for water, aluminum, and lead, respectively

1. The crossover between the two modes occurs at a critical kinetic energy $(E_{\text{K}})_{\text{crit}}$ where the two stopping powers are equal, i.e., $S_{\text{rad}}[(E_{\text{K}})_{\text{crit}}] = S_{\text{col}}[(E_{\text{K}})_{\text{crit}}]$ for a given absorber with atomic number Z
2. The critical kinetic energy $(E_{\text{K}})_{\text{crit}}$ can be estimated from the following relationship

$$(E_{\text{K}})_{\text{crit}} \approx \frac{800\ \text{MeV}}{Z}, \tag{6.64}$$

and for water, aluminum and lead it amounts to $\sim$100 MeV, $\sim$61 MeV and $\sim$10 MeV, respectively.
3. For high Z absorbers the dominance of radiation loss over collision loss starts at lower kinetic energies than in low Z absorbers. However, even in high Z media, such as lead and uranium, $(E_{\text{K}})_{\text{crit}}$ is at $\sim$10 MeV, well in the relativistic region.
4. The ratio of collision to radiation stopping power $(S_{\text{col}}/S_{\text{rad}})$ at a given electron kinetic energy may be estimated from the following

$$\frac{S_{\text{col}}}{S_{\text{rad}}} = \frac{800\ \text{MeV}}{Z\,E_{\text{K}}} = \frac{(E_{\text{K}})_{\text{crit}}}{E_{\text{K}}}. \tag{6.65}$$

Figure 6.11 shows the total mass stopping power S_{tot} for electrons (heavy solid curves) in water, aluminum and lead against the electron kinetic energy E_{K}. For comparison the radiation and collision components of the total stopping power of Figs. 6.2 and 6.10 are also shown.

6.7 Radiation Yield

The radiation yield (sometimes also referred to as the bremstrahlung yield) $Y[(E_K)_0]$ of a light charged particle with initial kinetic energy $(E_K)_0$ striking an absorber is defined as that fraction of the initial kinetic energy that is emitted as radiation with energy E_{rad} through the slowing down process of the particle in the absorber. The following features should be noted:

1. The energy E_{rad} is generally emitted in the form of bremsstrahlung, but can also be from positron in-flight annihilation, and in the form of characteristic radiation following impact ionization and impact excitation atomic events.
2. For heavy charged particles $Y\left[(E_K)_0\right] \approx 0$.
3. For light charged particles (electrons and positrons) the radiation yield $Y[(E_K)_0]$ is determined from stopping power data as follows:

$$Y[(E_K)_0] = \frac{\int\limits_0^{(E_K)_0} \frac{S_{rad}(E)}{S_{tot}(E)} dE}{\int\limits_0^{(E_K)_0} dE} = \frac{1}{(E_K)_0} \int\limits_0^{(E_K)_0} \frac{S_{rad}(E)}{S_{tot}(E)} dE. \tag{6.66}$$

4. For positron interactions, in-flight annihilation may also produce photons before the positron energy has been completely expended, so that in general the radiation yield is the sum of two contributions: the bremsstrahlung yield $Y_B[(E_K)_0]$ and the in-flight annihilation yield $Y_A[(E_K)_0]$. However, since the in-flight annihilation yield is much smaller than the bremsstrahlung yield, it is generally ignored in calculation of the radiation yield $Y[(E_K)_0]$ and the term "bremsstrahlung yield" is often incorrectly used to describe the total radiation yield.

Energy E_{rad} radiated per charged particle is from (6.66) given as

$$E_{rad} = (E_K)_0 \cdot Y[(E_K)_0] = \int\limits_0^{(E_K)_0} \frac{S_{rad}(E)}{S_{tot}(E)} dE, \tag{6.67}$$

while energy E_{col} lost through ionization per charged particle is

$$E_{col} = (E_K)_0 - E_{rad} = (E_K)_0 \cdot \{1 - Y[(E_K)_0]\} = \int\limits_0^{(E_K)_0} \frac{S_{col}(E)}{S_{tot}(E)} dE. \tag{6.68}$$

The radiation yield $Y[(E_K)_0]$ for various absorbing media from carbon at low Z to uranium at high Z is plotted against incident electron kinetic energy $(E_K)_0$ in Fig. 6.12. For a given incident electron kinetic energy $(E_K)_0$, the radiation yield $Y[(E_K)_0]$ increases with absorber atomic number Z and, for a given absorber atomic number Z, the radiation yield $Y[(E_K)_0]$ increases with increasing incident (initial) electron kinetic energy $(E_K)_0$.

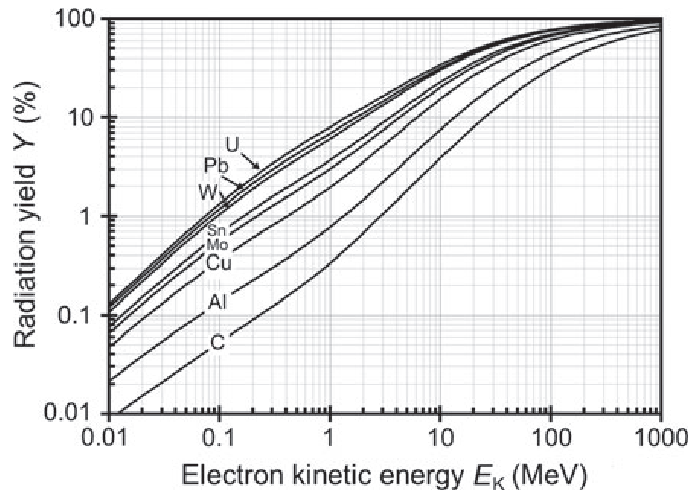


Fig. 6.12 Radiation yield $Y\left[(E_K)_0\right]$ in percent of incident charged particle kinetic energy $(E_K)_0$ for selected absorbing media from carbon to uranium against incident electron kinetic energy $(E_K)_0$. Data were obtained from the NIST

6.8 Range of Charged Particles

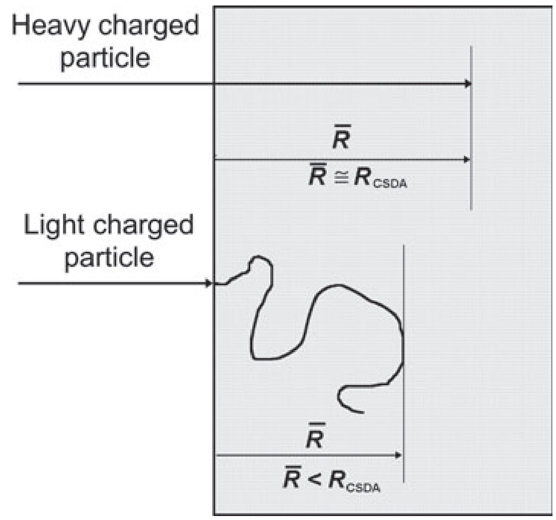
The range R of a charged particle in a particular absorbing medium is an experimental concept providing the thickness of an absorber that the particle can just penetrate. It depends on the particle's kinetic energy, mass as well as charge, and on the composition of the absorbing medium. In traversing matter charged particles lose their energy in ionizing and radiation collisions that may also result in significant deflections from their incident trajectory. In addition, charged particles suffer a large number of deflections as a result of elastic scattering.

These scattering effects are much more pronounced for light charged particles (electrons and positrons) in comparison to heavy charged particles because:

1. Heavy charged particles do not experience radiation losses, transfer only small amounts of energy in individual ionizing collisions, and mainly suffer small angle deflections in elastic collisions. Their path through an absorbing medium is thus essentially rectilinear, as shown schematically in Fig. 6.13.
2. Electrons with kinetic energy E_K, on the other hand, can lose energy up to $\frac{1}{2}E_K$ in individual ionizing collisions and energy up to E_K in individual radiation collisions. Since they can also be scattered with very large scattering angles, their path through the absorbing medium is very tortuous, as shown schematically in Fig. 6.13.

Various definitions of range are in common use. For example, the *path length* of a charged particle is the total distance along the particle's actual trajectory until it comes to rest, regardless of the direction of motion. The *projected range*, on the other hand, is the sum of individual path lengths projected onto the incident particle direction. Also used and defined below are the CSDA range R_{CSDA}, maximum range R_{max}, the 50% range R_{50}, practical range R_p, and the therapeutic ranges R_{80} and R_{90}.

Fig. 6.13 Schematic diagram of charged particle penetration into absorbing medium. *Top* Heavy charged particle (proton, α-particle, heavy ion, etc.); *bottom* light charged particle (electron or positron)



6.8.1 CSDA Range

Most of the collision and radiation interactions individually transfer only minute fractions of the incident particle's kinetic energy, and it is convenient to think of the particle that is moving through an absorber as losing its kinetic energy gradually and continuously in a process often referred to as the "continuous slowing down approximation" (CSDA). *Martin Berger* and *Stephen Seltzer* introduced the CSDA range concept in 1983 and defined the CSDA range R_{CSDA} as

$$R_{\mathrm{CSDA}} = \int_{0}^{(E_{\mathrm{K}})_0} \frac{\mathrm{d}E}{S_{\mathrm{tot}}(E)}, \tag{6.69}$$

where

R_{CSDA} is the CSDA range (mean path length) of the charged particle in the absorber (typically in $\mathrm{cm}^2 \cdot \mathrm{g}^{-1}$).
$(E_{\mathrm{K}})_0$ is the initial kinetic energy of the charged particle.
$S_{\mathrm{tot}}(E)$ is the total mass stopping power of the charged particle as a function of the kinetic energy E_{K}.

The CSDA range is a calculated quantity that represents the mean path length along the particle's trajectory and not necessarily the depth of penetration in a defined direction in the absorbing medium. For heavy charged particles, R_{CSDA} is a very good approximation to the average range $\overline{R}$ of the charged particle in the absorbing medium, because of the essentially rectilinear path of the charged particle (see Fig. 6.13) in the absorbing medium; for light charged particles the CSDA range can be up to twice the average range $\overline{R}$.

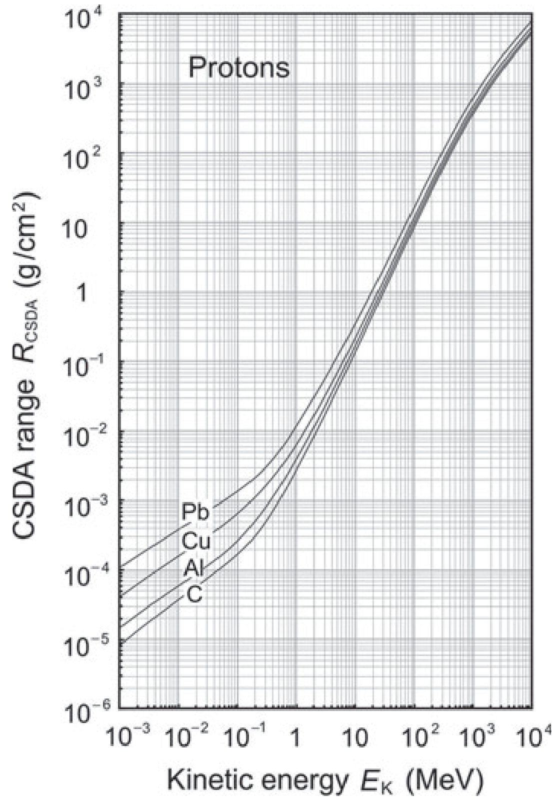


Fig. 6.14 CSDA range of protons against incident kinetic energy $(E_K)_0$ in carbon, aluminum, copper, and lead. Data are from the NIST

6.8.2 Maximum Penetration Depth

The maximum penetration depth R_{max} is defined as the depth in the absorbing medium beyond which no particles are observed to penetrate. For heavy charged particles $R_{max} \approx R_{CSDA}$ for all absorbing media, while for light charged particles $R_{max}/R_{CSDA} \approx 1$ for low Z absorbers and decreases with increasing Z to reach $R_{max}/R_{CSDA} \approx 0.5$ for high Z absorbers.

6.8.3 Range of Heavy Charged Particles in Absorbing Medium

Figure 6.14 gives the CSDA range R_{CSDA} of protons in aluminum, copper, and lead. Because of the decrease in the collision stopping power S_{col} with increasing atomic number Z of the absorber, for a given proton kinetic energy E_K, the range in g/cm^2 increases with the absorber atomic number Z. The CSDA range of heavy charged particles in an absorbing medium can be determined from the following expression

$$R_{\text{CSDA}} = \int\limits_0^{(E_\text{K})_0} \frac{\text{d}E_\text{K}}{S_{\text{col}}(E_\text{K})}, \tag{6.70}$$

after inserting into (6.69) the relationship $S_{\text{tot}} = S_{\text{col}}$, since for heavy charged particles S_{rad} is negligible. $(E_\text{K})_0$ in (6.70) is the initial kinetic energy of the incident heavy charged particle. The collision stopping power S_{col} for non-relativistic heavy charged particles is given quantum mechanically by the Bethe equation (6.55) as

$$S_{\text{col}} = C_0 \frac{z^2}{\beta^2} \frac{Z}{A} B_{\text{col}} = C_1 \frac{z^2}{\upsilon^2} \ln \frac{2m_\text{e}\upsilon^2}{I}, \tag{6.71}$$

where

- C_1 is a constant for a given absorber containing C_0 defined in (6.43) as well as the atomic number Z and atomic mass A of the absorber: $C_1 = C_0 c^2 Z/A$.
- B_{col} is the atomic stopping number defined in (6.53).
- υ is the velocity of the heavy charged particle.
- m_e is the rest mass of the electron.

For a non-relativistic heavy charged particle of mass M_0, its kinetic energy E_K and velocity υ are related through the classical expression $E_\text{K} = \frac{1}{2} M_0 \upsilon^2$. The CSDA range is then given as

$$R_{\text{CSDA}}^{(M_0)} = \frac{1}{C_1 z^2} \int\limits_0^{(E_\text{K})_0} \frac{\upsilon^2 \text{d}E}{\ln \dfrac{2m_\text{e}\upsilon^2}{I}} = \frac{2M_0}{C_1 z^2} \int\limits_0^{(E_\text{K})_0} \frac{\dfrac{E_\text{K}}{M_0} \text{d}\left(\dfrac{E_\text{K}}{M_0}\right)}{\ln\left[C_2 \dfrac{E_\text{K}}{M_0}\right]}, \tag{6.72}$$

where C_2 is a constant for a given absorbing medium: $C_2 = 4m_\text{e}/I$. For non-relativistic protons with rest mass m_p and incident kinetic energy $(E_\text{K})_0$, based on (6.72), the CSDA range in a given absorber is given as

$$R_{\text{CSDA}}^{(\text{p})}\left[(E_\text{K})_0\right] = \frac{2m_\text{p}}{C_1} \int\limits_0^{(E_\text{K})_0} \frac{\dfrac{E_\text{K}}{m_\text{p}} \text{d}\left(\dfrac{E_\text{K}}{m_\text{p}}\right)}{\ln\left[C_2 \dfrac{E_\text{K}}{m_\text{p}}\right]}. \tag{6.73}$$

The CSDA range for protons in various absorbers has been measured extensively in the past and the data are readily available in tabular as well as graphic form. It turns out that the proton CSDA range $\left(R_{\text{CSDA}}^{(\text{p})}\right)$ data can be used to estimate the CSDA range R_{CSDA} of any heavy charged particle following a simple relationship that links the two CSDA ranges.

The CSDA range R_{CSDA} (6.72) of an arbitrary heavy charged particle in a given absorber can be written in terms of the proton CSDA range as follows

$$
\begin{aligned}
R_{\mathrm{CSDA}}^{(M_0)}\left[(E_\mathrm{K})_0\right] &= \left(\frac{M_0}{m_\mathrm{p}}\right)\frac{2m_\mathrm{p}}{C_1 z^2}\int\limits_0^{(E_\mathrm{K})_0}\frac{\left(\dfrac{E_\mathrm{K}}{m_\mathrm{p}}\dfrac{m_\mathrm{p}}{M_0}\right)\mathrm{d}\left(\dfrac{E_\mathrm{K}}{m_\mathrm{p}}\dfrac{m_\mathrm{p}}{M_0}\right)}{\ln\left[C_2\dfrac{E_\mathrm{K}}{m_\mathrm{p}}\dfrac{m_\mathrm{p}}{M_0}\right]} \\
&= \frac{1}{z^2}\left(\frac{M_0}{m_p}\right)\left\{\frac{2m_p}{C_1}\int\limits_0^{(E'_\mathrm{K})_0}\frac{\dfrac{E'_\mathrm{K}}{m_\mathrm{p}}\,\mathrm{d}\left(\dfrac{E'_\mathrm{K}}{m_\mathrm{p}}\right)}{\ln\left[C_2\dfrac{E'_\mathrm{K}}{m_\mathrm{p}}\right]}\right\} \\
&= \frac{1}{z^2}\left(\frac{M_0}{m_\mathrm{p}}\right)R_{\mathrm{CSDA}}^{(\mathrm{p})}[\left(E'_\mathrm{K}\right)_0],
\end{aligned}
\tag{6.74}
$$

where we use the substitution

$$
E'_\mathrm{K} = E_\mathrm{K}\frac{m_\mathrm{p}}{M_0} \tag{6.75}
$$

and

$(E_\mathrm{K})_0$ is the incident kinetic energy of the arbitrary heavy charged particle.
$\left(E'_\mathrm{K}\right)_0$ is the corresponding incident kinetic energy of the proton given as

$$
\left(E'_\mathrm{K}\right)_0 = (E_\mathrm{K})_0\,\frac{m_\mathrm{p}}{M_0}. \tag{6.76}
$$

Therefore, the general procedure for finding R_{CSDA} of a heavy charged particle of rest mass M_0 and incident kinetic energy $(E_\mathrm{K})_0$ is to find in range tables for protons the $R_{\mathrm{CSDA}}^{(\mathrm{p})}$ at corresponding incident proton kinetic energy of $\left(E'_\mathrm{K}\right)_0 = (E_\mathrm{K})_0\,m_\mathrm{p}/M_0$ and then multiplying this value with $M_0/(m_\mathrm{p}z^2)$, as shown in (6.74).

For example, in a given absorber the CSDA range of a 20 MeV α-particle is the same as the range of a 5 MeV proton, since, according to (6.74), for the α-particle

$$
\frac{M_\alpha}{m_\mathrm{p}}\frac{1}{z^2} \approx 1 \qquad \text{and} \qquad \left(E'_\mathrm{K}\right)_0 = \frac{m_\mathrm{p}}{M_\alpha}\,(E_\mathrm{K})_0 \approx \frac{1}{4}\,(E_\mathrm{K})_0\,. \tag{6.77}
$$

Thus

$$
R_{\mathrm{CSDA}}^{(\alpha)}\,(20\ \mathrm{MeV}) \approx R_{\mathrm{CSDA}}^{(\mathrm{p})}\,(5\ \mathrm{MeV})\,. \tag{6.78}
$$

Another example: a deuteron with kinetic energy of 20 MeV has the same velocity and stopping power as a 10 MeV proton. However, its CSDA range is twice that of a 10 MeV proton, since

$$\frac{M_{\rm d}}{m_{\rm p}}\frac{1}{z^2} \approx 2 \qquad \text{and} \qquad \left(E'_{\rm K}\right)_0 = (E_{\rm K})_0\,\frac{m_{\rm p}}{M_{\rm d}} \approx \frac{1}{2}\,(E_{\rm K})_0\,. \tag{6.79}$$

Thus

$$R^{\rm (d)}_{\rm CSDA}\,(20\ {\rm MeV}) \approx 2R^{\rm (p)}_{\rm CSDA}\,(10\ {\rm MeV})\,. \tag{6.80}$$

6.8.4 Range of Light Charged Particles (Electrons and Positrons) in Absorbers

For light charged particles, the CSDA range $R_{\rm CSDA}$ exceeds the average range $\overline{R}$ in an absorbing medium, because of the very tortuous path that the light charged particles experience in the absorbing medium (see Fig. 6.13). For low atomic number absorbers the difference is only about 10% to 15%, however, for high Z absorbers, the CSDA range can be up to twice the average range of charged particles in the absorber.

Electrons are used in external beam radiotherapy for treatment of superficial lesions, therefore, accurate knowledge of electron range in water and tissue is important. Since the CSDA range can serve only as a rough guide on the penetration of electron beams into tissue, other more appropriate ranges have been defined for use in radiotherapy, all of them based on measurement of electron depth dose distribution in water.

A typical electron beam depth dose curve (dose versus depth in water, see Sect. 1.12.3) is plotted in Fig. 6.15. The dose distribution is normalized to 100% at the depth of dose maximum $z_{\rm max}$, and exhibits a relatively high surface dose, a rapid dose fall off beyond $z_{\rm max}$, and a leveling off of the dose at a low level component referred to as the bremsstrahlung tail. Various electron ranges of interest in radiotherapy and radiation dosimetry, such as R_{80}, R_{90}, $R_{\rm p}$, and $R_{\rm max}$, are identified on the depth dose curve of Fig. 6.15.

The maximum range $R_{\rm max}$ is defined as the depth at which extrapolation of the tail of the depth dose curve meets the bremsstrahlung background. It is the largest penetration depth of electrons in the absorbing medium. The drawback of $R_{\rm max}$ is that it does not provide a well defined measurement point.

The practical range $R_{\rm p}$ is defined as the depth at which the tangent plotted through the steepest section of the electron depth dose curve intersects with the extrapolation line of the bremsstrahlung background. It is used for the determination of $\overline{E}_{\rm K}(z)$, the mean electron beam kinetic energy at a depth z in a water phantom, with a relationship proposed by Harder

$$\overline{E}_{\rm K}(z) = \overline{E}_{\rm K}(0)\left[1 - \frac{z}{R_{\rm p}}\right], \tag{6.81}$$

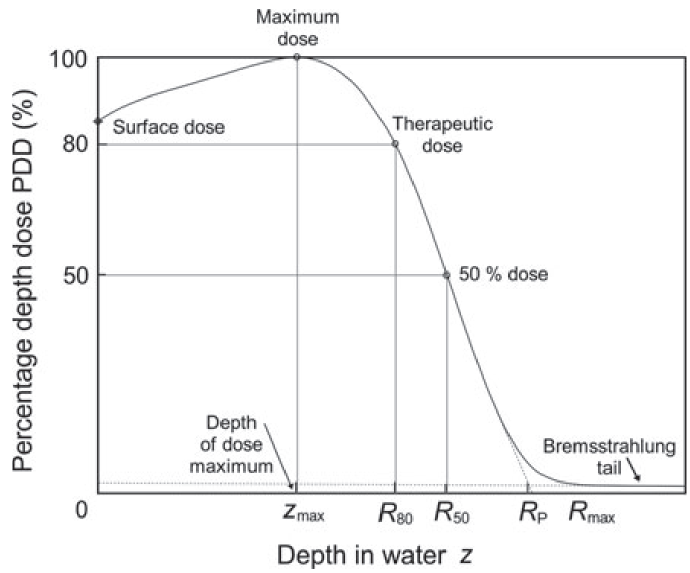


Fig. 6.15 Typical electron beam depth dose curve (dose against depth in water) normalized to 100 at the depth of dose maximum z_{max}. Several ranges of interest in radiotherapy and dosimetry, such as R_{80}, R_{50}, R_p, and R_{max}, are identified on the curve

where $\overline{E}_K(0)$ is the mean electron kinetic energy at the water phantom surface at $z = 0$.

The depths R_{90}, R_{80}, and R_{50} on the depth dose curve are defined as depths at which the percentage depth doses beyond the depth of dose maximum z_{max} attain values of 90%, 80%, and 50%, respectively. R_{90} or R_{80} are used for prescription of tumor dose in radiotherapy, while R_{50} is used in radiation dosimetry for determination of the mean electron kinetic energy on the water surface $\overline{E}_K(0)$ which is then in turn used in the Harder relationship of (6.81). The relationship between R_{50} and $\overline{E}_K(0)$ is as follows

$$\overline{E}_K(0) = C R_{50}, \tag{6.82}$$

where C is a constant, for water equal to 2.33 MeV/cm.

6.9 Mean Collision Stopping Power

In radiation dosimetry the main interest is in the energy absorbed per unit mass of the absorbing medium governed by collision losses of charged particles. It is often convenient to characterize a given radiation beam with electrons of only one energy rather than with an electron spectrum $d\phi/dE$ that is present in practice. For example, monoenergetic electrons set in motion with an initial kinetic energy $(E_K)_0$ in an absorbing medium will through their own slowing down process produce a spectrum of electrons in the medium ranging in energy from $(E_K)_0$ down to zero.

The electron spectrum $\mathrm{d}\phi(E)/\mathrm{d}E$, when ignoring any possible hard collisions, is given as

$$\frac{\mathrm{d}\phi(E)}{\mathrm{d}E} = \frac{N}{S_{\mathrm{tot}}(E)}, \tag{6.83}$$

where

N is the number of monoenergetic electrons of kinetic energy $(E_{\mathrm{K}})_0$ produced per unit mass in the absorbing medium.
$S_{\mathrm{tot}}(E)$ is the total stopping power.

For this electron spectrum, produced by monoenergetic electrons, as shown by Harold E. Johns and John R. Cunningham, we can define a *mean collision stopping power* $\overline{S}_{\mathrm{col}}$ as follows

$$\overline{S}_{\mathrm{col}}\left[(E_{\mathrm{K}})_0\right] = \frac{\int_0^{(E_{\mathrm{K}})_0} \frac{\mathrm{d}\phi}{\mathrm{d}E} S_{\mathrm{col}}(E)\mathrm{d}E}{\int_0^{(E_{\mathrm{K}})_0} \frac{\mathrm{d}\phi}{\mathrm{d}E}\mathrm{d}E}. \tag{6.84}$$

Using (6.69) and (6.83) the integral in the denominator of (6.84) is determined as follows

$$\int_0^{(E_{\mathrm{K}})_0} \frac{\mathrm{d}\phi}{\mathrm{d}E}\mathrm{d}E = N \int_0^{(E_{\mathrm{K}})_0} \frac{\mathrm{d}E}{S_{\mathrm{tot}}(E)} = N \cdot R_{\mathrm{CSDA}}. \tag{6.85}$$

Using (6.63), (6.66) and (6.83), the numerator of (6.86) is determined as follows

$$\begin{aligned}\int_0^{(E_{\mathrm{K}})_0} \frac{\mathrm{d}\phi}{\mathrm{d}E} S_{\mathrm{col}}(E)\mathrm{d}E &= N \int_0^{(E_{\mathrm{K}})_0} \frac{S_{\mathrm{col}}(E)}{S_{\mathrm{tot}}(E)}\,\mathrm{d}E \\ &= N \int_0^{(E_{\mathrm{K}})_0} \frac{S_{\mathrm{tot}}(E) - S_{\mathrm{rad}}(E)}{S_{\mathrm{tot}}(E)}\,\mathrm{d}E = N\cdot(E_{\mathrm{K}})_0 - N\cdot(E_{\mathrm{K}})_0 \cdot Y[(E_{\mathrm{K}})_0].\end{aligned} \tag{6.86}$$

The mean collision stopping power $\overline{S}_{\mathrm{col}}\left[(E_{\mathrm{K}})_0\right]$ of (6.84) can now be written as

$$\overline{S}_{\mathrm{col}}[(E_{\mathrm{K}})_0] = (E_{\mathrm{K}})_0 \frac{1 - Y[(E_{\mathrm{K}})_0]}{R_{\mathrm{CSDA}}}. \tag{6.87}$$

The relationship (6.87) for $\overline{S}_{\mathrm{col}}\,[(E_{\mathrm{K}})_0]$ could also be stated intuitively by noting that an electron with an initial kinetic energy $(E_{\mathrm{K}})_0$ will, through traveling the pathlength ℓ equal to R_{CSDA} in the absorbing medium: (1) lose an energy $(E_{\mathrm{K}})_0\,Y[(E_{\mathrm{K}})_0]$ to bremsstrahlung and (2) deposit an energy $(E_K)_0 \{1 - Y[(E_K)_0]\}$ in the medium. The same would hold for a positron except that it could also lose part of its energy to photon production in in-flight annihilation events.

6.10 Restricted Collision Stopping Power

In radiation dosimetry one is interested in determining the energy transferred to a localized region of interest; however, the use of the mass collision stopping power S_{col} for this purpose may overestimate the dose because S_{col} incorporates both hard and soft collisions. The δ-rays resulting from hard collisions may be energetic enough to carry their kinetic energy a significant distance from the track of the primary particle thereby escaping from the region of interest in which the dose is determined.

The concept of restricted mass collision stopping power L_Δ has been introduced to address this issue by excluding the δ-rays with energies exceeding a suitable threshold value Δ.

The choice of the energy threshold Δ depends on the problem at hand. For dosimetric measurements involving air-filled ionization chambers with a typical electrode separation of 2 mm a frequently used threshold value is 10 keV (Note: the range of a 10 keV electron in air is of the order of 2 mm). For microdosimetric studies, on the other hand, one usually takes 100 eV as a reasonable threshold Δ value.

Of course, to be physically relevant Δ must not exceed ΔE_{max}, the maximum possible energy transfer to orbital electron from the incident particle with kinetic energy E_K in a direct-hit collision. As shown in Sect. 5.3.6, ΔE_{max} equals to $\frac{1}{2}E_K$ for electrons, E_K for positrons, and $2m_ec^2\beta^2/(1-\beta^2)$ for heavy charged particles [see (5.46), (5.45), and (5.41), respectively].

For a given kinetic energy E_K of the primary particle the restricted collision stopping power L_Δ is in general smaller than the unrestricted collision stopping power S_{col}; the smaller is the threshold Δ, the larger is the discrepancy. As Δ increases from a very small value, the discrepancy diminishes until at $\Delta = \Delta E_{max}$ the restricted and unrestricted collision stopping powers become equal, i.e., $L_{\Delta=\Delta E_{max}} = S_{col}$, irrespective of E_K.

Figure 6.16 displays the unrestricted collision mass stopping power as well as the restricted collision mass stopping powers with $\Delta = 10$ keV and $\Delta = 100$ keV against kinetic energy E_K for electrons in carbon based on data in the ICRU Report 37. The following observations can now be made:

1. Since energy transfers to secondary electrons are limited to $\frac{1}{2}E_K$, the unrestricted mass collision stopping power S_{col} and restricted mass collision stopping power L_Δ are identical for a given kinetic energy E_K of the electron for kinetic energies lower than or equal to 2Δ. This is indicated in Fig. 6.16 with vertical lines at 20 keV and 200 keV for the threshold values $\Delta = 10$ keV and $\Delta = 100$ keV, respectively.

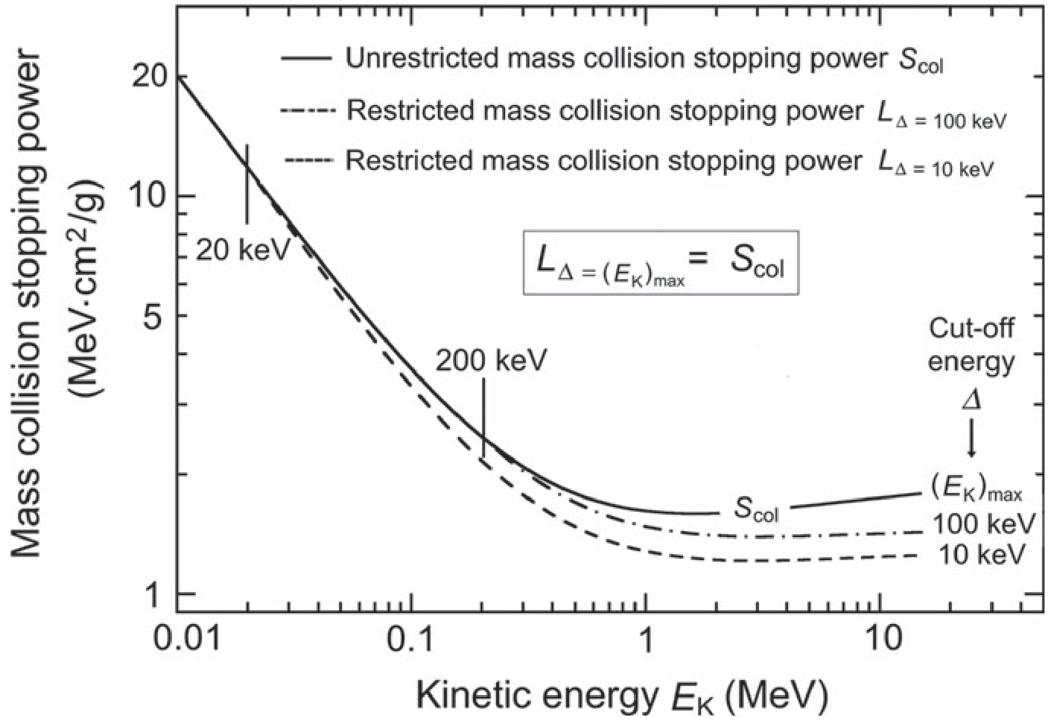


Fig. 6.16 Unrestricted mass collision stopping power S_{col} and restricted mass collision stopping power L_Δ with thresholds $\Delta = 10$ keV and $\Delta = 100$ keV for electrons in carbon against kinetic energy E_K. Data are based on the ICRU Report 37

2. For a given $E_K > 2\Delta$, the smaller is Δ, the larger is the discrepancy between the unrestricted and restricted stopping powers.
3. For given Δ and $E_K > 2\Delta$, the larger is E_K, the larger is the discrepancy between the unrestricted and restricted stopping powers.

6.11 Bremsstrahlung Targets

Bremsstrahlung production is of great importance in medical physics, since the majority of radiation beams used in diagnostic radiology and in external beam radiotherapy are produced through bremsstrahlung interactions of monoenergetic electrons with solid targets. These targets are components of x-ray machines and linear accelerators; the most commonly used radiation-emitting machines for diagnosis and treatment of disease.

An electron that strikes the target with a given kinetic energy will undergo many different interactions with target atoms before it comes to rest and dissipates all of its kinetic energy in the target. As discussed in Sect. 6.1, there are two classes of electron interactions with a target atom:

1. Incident electron interaction with orbital electron of a target atom results mainly in collision impact loss and ionization of the target atom that may be accompanied by an energetic electron referred to as delta ray. The collision loss in an x-ray target is followed by emission of characteristic x-rays and Auger electrons.
2. Incident electron interaction with the nucleus of a target atom results mainly in elastic scattering events but may also result in radiation loss accompanied with bremsstrahlung production.

While bremsstrahlung is a major contributor to the x-ray spectrum at superficial (50 kVp to 100 kVp) and orthovoltage (100 kVp to 350 kVp) energies, it is essentially the sole contributor to the x-ray spectrum at megavoltage energies. With regard to their thickness compared to the average range $\overline{R}$ of electrons in the target material, x-ray targets are either *thin* or *thick*; the thickness of a thin target is much smaller than $\overline{R}$, while the thickness of a thick target is of the order of $\overline{R}$.

As discussed in Sect. 4.2.8, the peak x-ray intensity occurs at a characteristic angle θ_{max} that depends on the kinetic energy of incident electrons. As shown schematically in Fig. 6.17:

1. In the diagnostic energy range (50 kVp to 350 kVp) x-ray tubes are used for production of x-rays and most photons are emitted at 90° from the direction of electron deceleration in the target. The characteristic angle θ_{max} thus equals $\sim$90° (see Fig. 6.17a).
2. In the megavoltage radiotherapy range (4 MV and above) acceleration waveguides are used for electron acceleration and x-ray production and most photons are emitted in the direction of electron deceleration in the target. The characteristic angle θ_{max} is then $\sim$0° and the target is referred to as a transmission target (see Fig. 6.17b).

X-ray beams produced in x-ray targets are heterogeneous and contain photons of many energies ranging from 0 to a maximum energy $h\nu_{max}$ which is equal to the

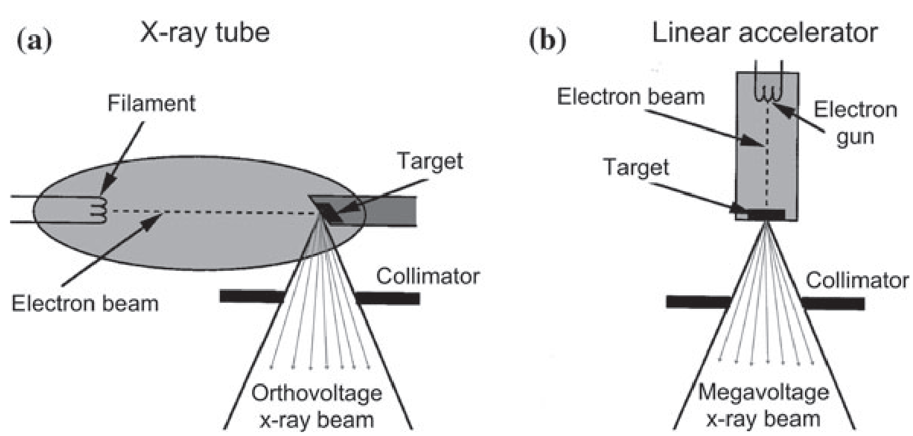


Fig. 6.17 Schematic comparison of x-ray production in the diagnostic radiology orthovoltage range with an x-ray machine (x-rays emitted mostly orthogonally to the electron beam) and in the radiotherapy megavoltage range with a linear accelerator (x-rays emitted mostly in the direction of the electron beam)

kinetic energy E_K of the electrons striking the target. The relationship

$$h\nu_\mathrm{max} = E_\mathrm{K} = eU \tag{6.88}$$

is referred to as the Duane–Hunt law in honor of two American physicists, *William Duane* and *Franklin Hunt*, who discovered the law in 1915. In its original form the Duane–Hunt law stated that there is a sharp upper limit to the x-ray frequencies emitted from an x-ray target stimulated by an impact of electrons and that this frequency ν_max is given by the quantum connection, independently of the target material, as

$$\nu_\mathrm{max} = \frac{c}{\lambda_\mathrm{min}} = \frac{eU}{h}, \tag{6.89}$$

where

c is the speed of light in vacuum.
λ_min is the minimum photon wavelength.
e is the electron charge.
U is the potential difference used in electron acceleration.

The Duane–Hunt law, sometimes called the inverse photoelectric effect, defines the maximum photon energy $h\nu_\mathrm{max}$ in an x-ray beam produced by thin or thick x-ray targets. However, to describe a heterogeneous x-ray beam we must provide information on a full x-ray spectrum, since the spectrum is affected not only by the maximum photon energy $h\nu_\mathrm{max}$ but also by the target type (thin or thick) and target atomic number Z.

The spectrum of an x-ray beam is most commonly shown by plotting either:

1. Number of photons per energy interval $[\Delta N/\Delta(h\nu)]$ against photon energy $h\nu$ of the given energy interval bin

or

2. Photon intensity I_ν against photon energy $h\nu$ of a given energy interval bin, where the photon intensity I_ν is proportional to the product $(\Delta N/\Delta E)\cdot h\nu$.

X-ray spectra can also be presented by plotting $(\Delta N/\Delta E)$ or beam intensity I against photon wavelength λ. In this case the Duane–Hunt law is given by specifying the minimum wavelength λ_min rather than the maximum frequency ν_max, where $\lambda_\mathrm{min} = c/\nu_\mathrm{max}$. Since $I_\nu \mathrm{d}\nu = I_\lambda \mathrm{d}\lambda$ the following relationship also applies

$$I_\lambda = I_\nu \left|\frac{\mathrm{d}\nu}{\mathrm{d}\lambda}\right| = \left|I_\nu \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{c}{\lambda}\right)\right| = \frac{c}{\lambda^2} I_\nu. \tag{6.90}$$

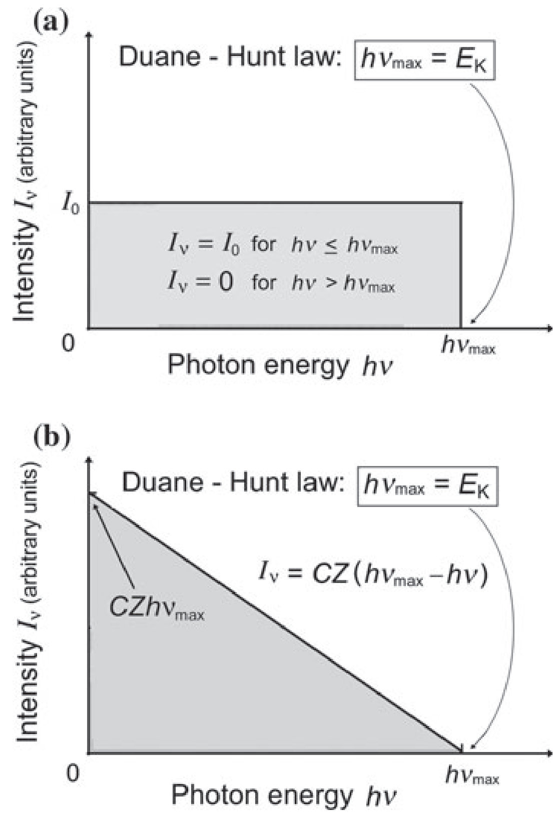


Fig. 6.18 Typical x-ray intensity spectrum: **a** for thin target radiation and **b** for thick target radiation

6.11.1 Thin X-Ray Targets

Thin x-ray targets are mainly of theoretical interest and their thickness is very small compared to the range of electrons of given kinetic energy in the target material. By definition, a thin target is so thin that electron striking it:

1. Loses essentially no energy by atomic ionizations.
2. Suffers no significant elastic collisions.
3. Traverses the target without interacting with target atoms or experiences only one bremsstrahlung interaction while traversing the target.

The bremsstrahlung radiation produced in a thin target by electrons of kinetic energy E_K has a constant intensity $I_\nu = I_0$ for $h\nu \le h\nu_{max}$ and zero intensity $I_\nu = 0$ for $h\nu > h\nu_{max}$, as shown schematically in Fig. 6.18a. The maximum photon energy in the spectrum follows the Duane–Hunt law (6.89). Since the x-ray intensity is proportional to the product $(\Delta N/\Delta h\nu) \cdot h\nu$, it follows that, in comparison with the number of photons of energy $h\nu_{max}$, the x-ray spectrum contains twice as many photons of energy $0.50\,h\nu_{max}$; 4 times as many photons of energy $0.25\,h\nu_{max}$; 10 times as many photons of energy $0.10\,h\nu_{max}$, etc.

6.11.2 Thick X-Ray Targets

Thick x-ray targets have thicknesses of the order of the average range of electrons $\overline{R}$ in the target material. In practice, typical thicknesses are equal to about $1.1\overline{R}$ to satisfy two opposing conditions:

1. To ensure that no electrons that strike the target can traverse the target.
2. To minimize the attenuation of the bremsstrahlung beam produced by electrons that are scattered (decelerated) many times in the target.

Thick target radiation is much more difficult to handle theoretically than thin target radiation; however, in practice most targets used in bremsstrahlung production are of the thick target variety. A typical spectrum of a clinical x-ray beam consists of spectral lines characteristic of the target as well as the filtration material and superimposed onto the continuous bremsstrahlung spectrum. The bremsstrahlung spectrum originates in the x-ray target, while the characteristic spectrum originates not only in the target but also in any filtration placed into the x-ray beam.

Figure 6.19a displays three typical spectra from a tungsten target:

1. Unfiltered 100 kVp bremsstrahlung spectrum with superimposed tungsten K_α and K_β lines.
2. Filtered 100 kVp bremsstrahlung spectrum with superimposed tungsten K_α and K_β lines.
3. Filtered 60 kVp bremsstrahlung spectrum. The K lines are not present in the spectrum because the ionization energy of the K shell in tungsten is 70 keV and the kinetic energy of electrons striking the tungsten target is only 60 keV.

The spectral distribution of thick-target bremsstrahlung can be represented as a superposition of contributions from a large number of thin targets, each thin target traversed by a lower energy monoenergetic electron beam having a lower $h\nu_{\max}$ than the previous thin target. This is shown schematically in Fig. 6.19b which depicts a typical thin target bremsstrahlung spectrum (curve 1) and several thick target bremsstrahlung spectra (curves 2, 3, and 4) for an x-ray tube in which 100 keV electrons strike the target. Curve 1 is for a thin target producing constant beam intensity for photon energies from 0 to the kinetic energy of the electrons striking the target (100 keV). Curve 2 represents an unfiltered spectrum (inside the x-ray tube) for a thick target and a superposition of numerous thin target spectra; curve 3 is spectrum for a beam filtered by an x-ray tube window which filters out the low energy photons; and curve 4 is spectrum for a beam filtered by the x-ray tube window and additional filtration.

As indicated in Figs. 6.18b and 6.19b, the beam intensity I_ν for a thick target as a function of photon energy $h\nu$ may be described with an approximate linear relationship

$$I_\nu \approx CZ\,(h\nu_{\max} - h\nu)\,, \tag{6.91}$$

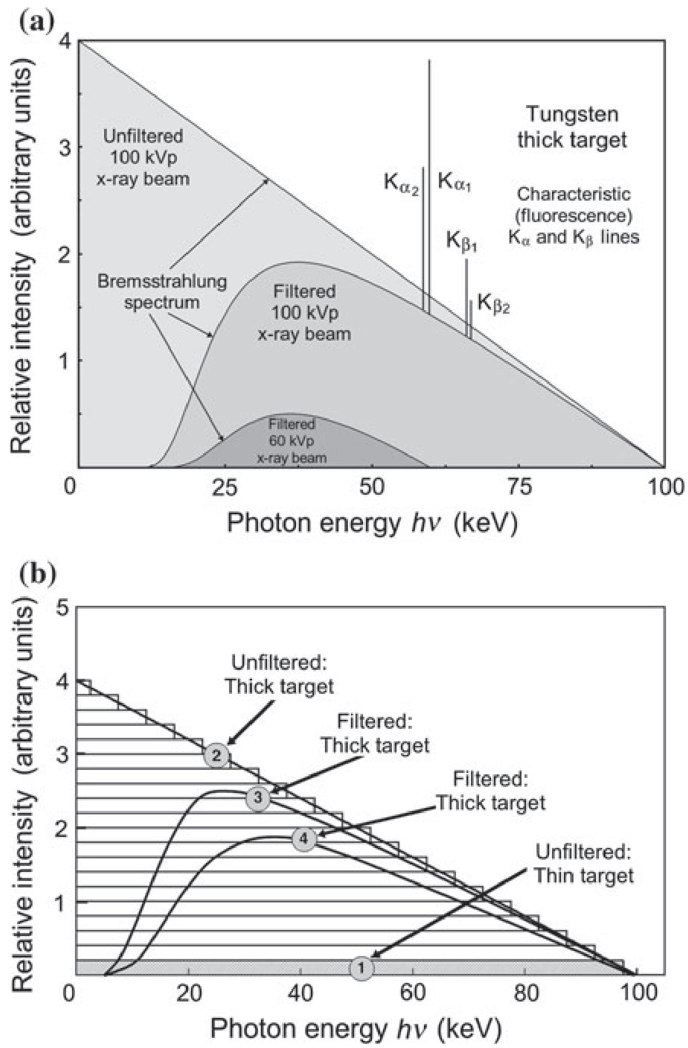


Fig. 6.19 Typical spectra produced by electrons with kinetic energy of 100 keV striking a tungsten target. *Part **a*** displays characteristic line spectra superimposed onto the bremsstrahlung spectrum and *part **b*** displays only the bremsstrahlung spectrum under various conditions: curve *1* is for a thin target, producing a constant intensity beam; curves *2*, *3*, and *4* are for a thick target with curve *2* representing an unfiltered spectrum (inside the x-ray tube), curve *3* representing an x-ray beam filtered by an x-ray tube window, and curve *4* representing an x-ray beam filtered by the x-ray tube window and additional filtration

where

C	is a constant.
Z	is the atomic number of the thick target.
$h\nu_{\max}$	is the Duane–Hunt photon energy limit.
I_ν	is the beam intensity at photon energy $h\nu$ with a maximum value $CZh\nu_{\max}$ at $h\nu = 0$ and a value of zero for $h\nu \geq h\nu_{\max}$.

The total intensity I is determined by integrating I_ν over the whole energy range from 0 to $h\nu_{\max}$

$$I = CZ \int_0^{h\nu_{\max}} (h\nu_{\max} - h\nu)\, \mathrm{d}h\nu = \frac{1}{2} CZ\, (h\nu_{\max})^2 = \frac{1}{2} CZ\, (eU)^2\,, \tag{6.92}$$

where U is the accelerating voltage and eU is the electron kinetic energy.

The total photon intensity I emitted from the x-ray target is thus proportional to:

1. Target atomic number Z.
2. Square of the accelerating potential U.

The average energy E_{rad} radiated by an electron of initial energy $(E_{\text{K}})_0$ in being stopped in a thick target was given in (6.67) as

$$E_{\text{rad}} = (E_{\text{K}})_0 \; Y\left[(E_{\text{K}})_0\right] = \int_0^{(E_{\text{K}})_0} \frac{S_{\text{rad}}(E)}{S_{\text{tot}}(E)} \text{d}E. \tag{6.93}$$

In the diagnostic energy range, where $(E_{\text{K}})_0 \ll m_{\text{e}}c^2$, the mass radiation stopping power S_{rad} is independent of the electron kinetic energy and, from (6.6) combined with Table 6.1, given as

$$S_{\text{rad}} = \frac{N_{\text{A}}}{A}\sigma_{\text{rad}} E_{\text{i}} = \frac{16}{3}\alpha\, r_{\text{e}}^2 Z^2 E_{\text{i}} = \frac{16}{3}\alpha\, r_{\text{e}}^2 \frac{N_{\text{A}}}{A} Z^2 \left[(E_{\text{K}})_0 + m_{\text{e}}c^2\right]. \tag{6.94}$$

For $(E_{\text{K}})_0 \ll m_{\text{e}}c^2$ the mass radiation stopping power S_{rad} of (6.93) is independent of the kinetic energy of the electron and (6.93) may then be simplified to read

$$E_{\text{rad}} = S_{\text{rad}} \int_0^{(E_{\text{K}})_0} \frac{\text{d}E}{S_{\text{tot}}(E)} = S_{\text{rad}} R_{\text{CSDA}} = \text{const}\frac{N_{\text{A}}}{A} Z^2 R_{\text{CSDA}}. \tag{6.95}$$

Since in the low energy range $S_{\text{tot}} \approx S_{\text{col}}$ and $S_{\text{col}} \propto N_{\text{A}} Z/A$, we note that $R_{\text{CSDA}} \propto (N_A Z/A)^{-1}$ and the average energy radiated by the electron stopped in a thick target is linearly proportional to the atomic number Z of the target, i.e.,

$$E_{\text{rad}} \propto Z\, f\left[(E_{\text{K}})_0\,, Z\right], \tag{6.96}$$

where $f\left[(E_{\text{K}})_0\,, Z\right]$ is a slowly varying function of the target atomic number Z.

A few other notable features of bremsstrahlung targets used in diagnostic and therapeutic medical equipment are listed below:

- The integrated intensity of thick-target bremsstrahlung depends linearly on the atomic number Z of the target material. This implies that high Z targets will be more efficient for x-ray production that low Z targets.
- In megavoltage radiotherapy only photons in the narrow cone in the forward direction are used for the clinical beams and the radiation yield in the forward direction is essentially independent of the atomic number Z of the target.
- The thick target bremsstrahlung is linearly proportional to the atomic number Z of the target in the diagnostic energy range where $(E_{\text{K}})_0 \ll m_{\text{e}}c^2$. This rule will fail when $(E_{\text{K}})_0$ becomes large enough for the radiation losses to no longer be negligible in comparison with collision losses.

Computerized Tomography Images and Leonardo Da Vinci

Center image on next page is a famous sketch of Vitruvian man by *Leonardo da Vinci* (1452–1519). The drawing is based on geometrical proportions of man proposed by the first century Roman architect Vitruvian, hence the name Vitruvian man. While other great men and women that humanity produced in arts and science generally excelled in only one specific area of art or science, Leonardo da Vinci was a man of enormous talents covering most areas of human endeavor, be it in arts or science; a truly versatile renaissance man. He was active as sculptor, painter, musician, architect, engineer, inventor, and researcher of the human body. Many of da Vinci's drawings of the human body helped doctors of his time to understand better the layout of muscle and bone structures within the body.

Left and right images on next page present *computerized tomography* (CT) images of the human body, representing the most important development that resulted from *Wilhelm Roentgen's* discovery of x-rays in 1895. A CT scanner is a machine that uses an x-ray beam rotating about a specific area of a patient to collect x-ray attenuation data for patient's tissues. It then manipulates these data with special mathematical algorithms to display a series of transverse slices through the patient. The transverse CT data can be reconstructed so as to obtain sagittal sections (shown in the right image on next page) and coronal sections (shown on the left image on next page) through the patient's organs or to obtain digitally reconstructed radiographs. The excellent resolution obtained with a modern CT scanner provides an extremely versatile "non-invasive" diagnostic tool. CT scanners have been in clinical and industrial use since the early 1970s and evolved through five generations, each generation increasingly more sophisticated and faster than the previous one.

Initially, CT scanners have been used for imaging in radiology departments and for target delineation in radiotherapy. Since the early 1990s CT scanners are also an invaluable tool in radiotherapy departments where they form part of a CT-simulator machine which allows not only visualization of internal organs and target delineation but also the so-called virtual simulation and actual treatment planning for radiotherapy.

Three types of detector are used in CT scanners: (1) scintillation detectors (sodium iodide or calcium fluoride) in conjunction with a photomultiplier tube; (2) gas (xenon or krypton) ionization chamber; and (3) semiconductor detectors (cesium iodide) in conjunction with a p–n junction photodiode.

Three-dimensional images can be obtained through three techniques: (1) multiple 2D acquisitions based on a series of sequential scans; (2) spiral (helical) CT with the x-ray source rotating continuously around the patient while simultaneously the patient is translated through the gantry; and (3) cone-beam CT using a 2D detector array in order to measure the entire volume-of-interest during one single orbit of the x-ray source.

Allan Cormack (1924–1998), a South African–American physicist, developed the theoretical foundations that made computerized tomography possible and published his work during 1963–64. His work generated little interest until *Godfrey Hounsfield* (1919–2004), a British electrical engineer, developed a practical model of a CT scanner in the early 1970s. Hounsfield and Cormack received the 1979 Nobel Prize in Physiology and Medicine for their independent invention of the CT scanner.